ELSEWEDY UNIVERSITY OF TECHNOLOGY
- POLYTECHNIC OF EGYPT -

Internet of Things

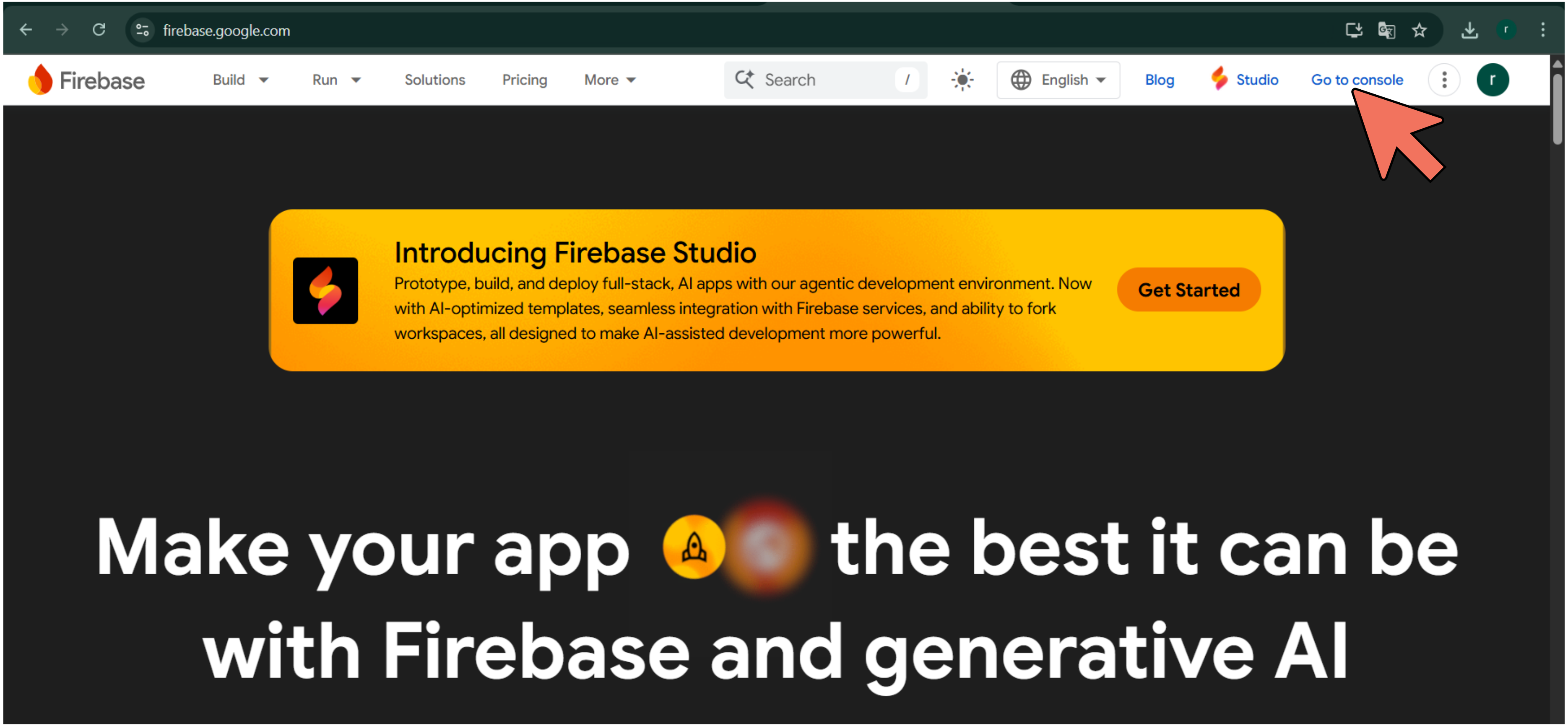
Lab 4

Firebase with IoT

- **Firebase** is Google's mobile and web application development platform that includes many services to manage data from IOS, Android, or web applications.
- This includes things like analytics, authentication, databases, configuration, file storage, push messaging, and the list goes on.
- The services are hosted in the cloud and scale with little to no effort on the part of the developer.
- You can use the ESP32 to connect and interact with your Firebase project, and you can create applications to control the ESP32 via Firebase from anywhere in the world.

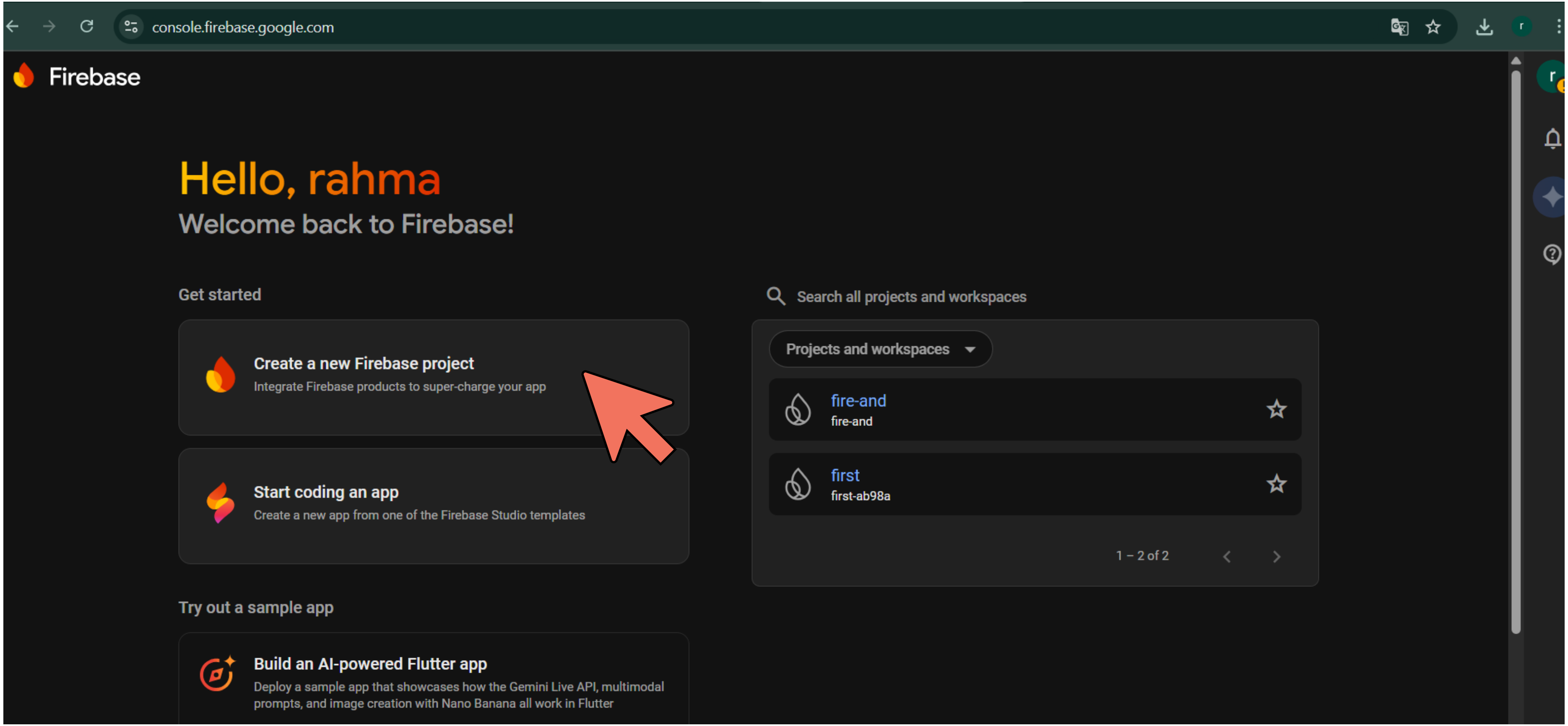
Firebase: Creating a New Project

❖ Go to https://firebase.google.com ,and then click on Go to console



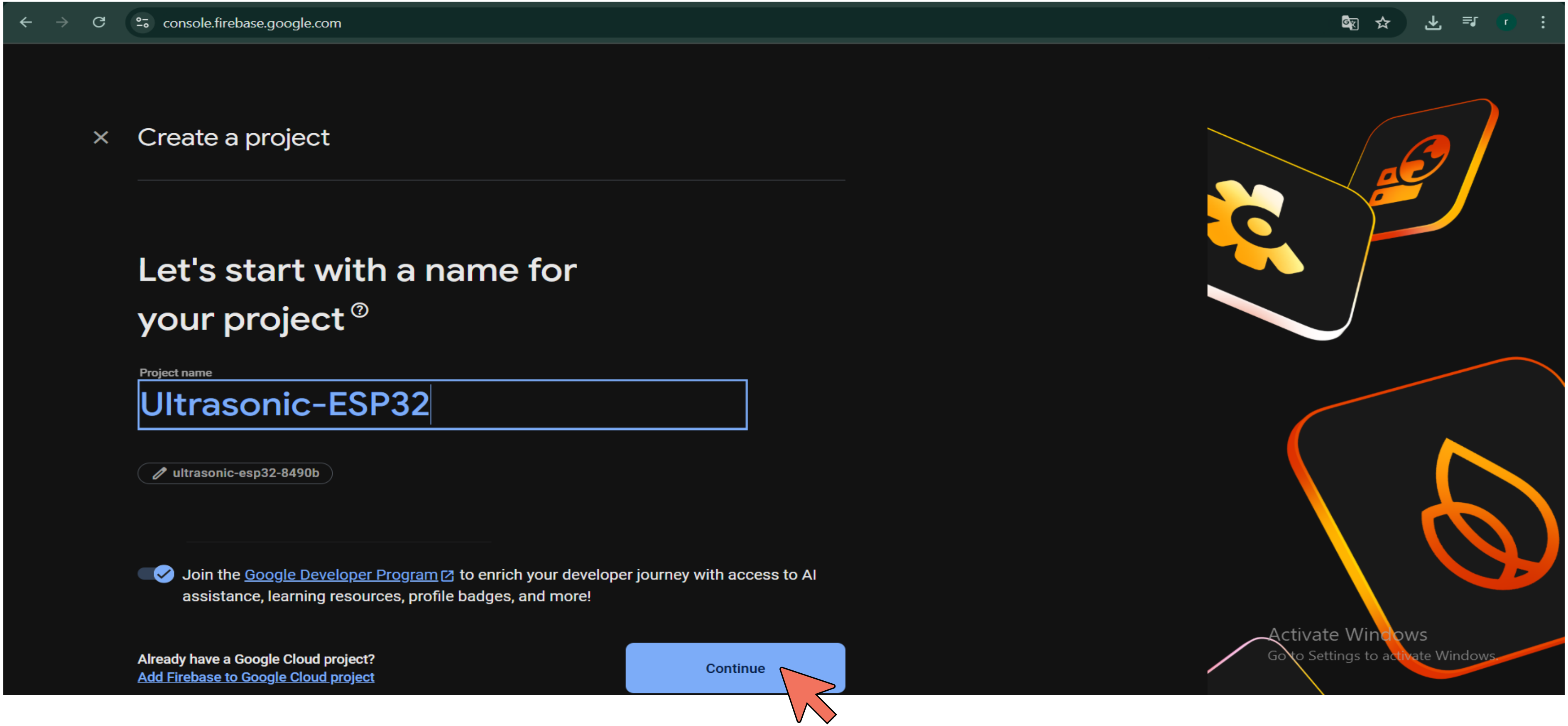
Firebase: Creating a New Project

❖ Click on create a new firebase project.



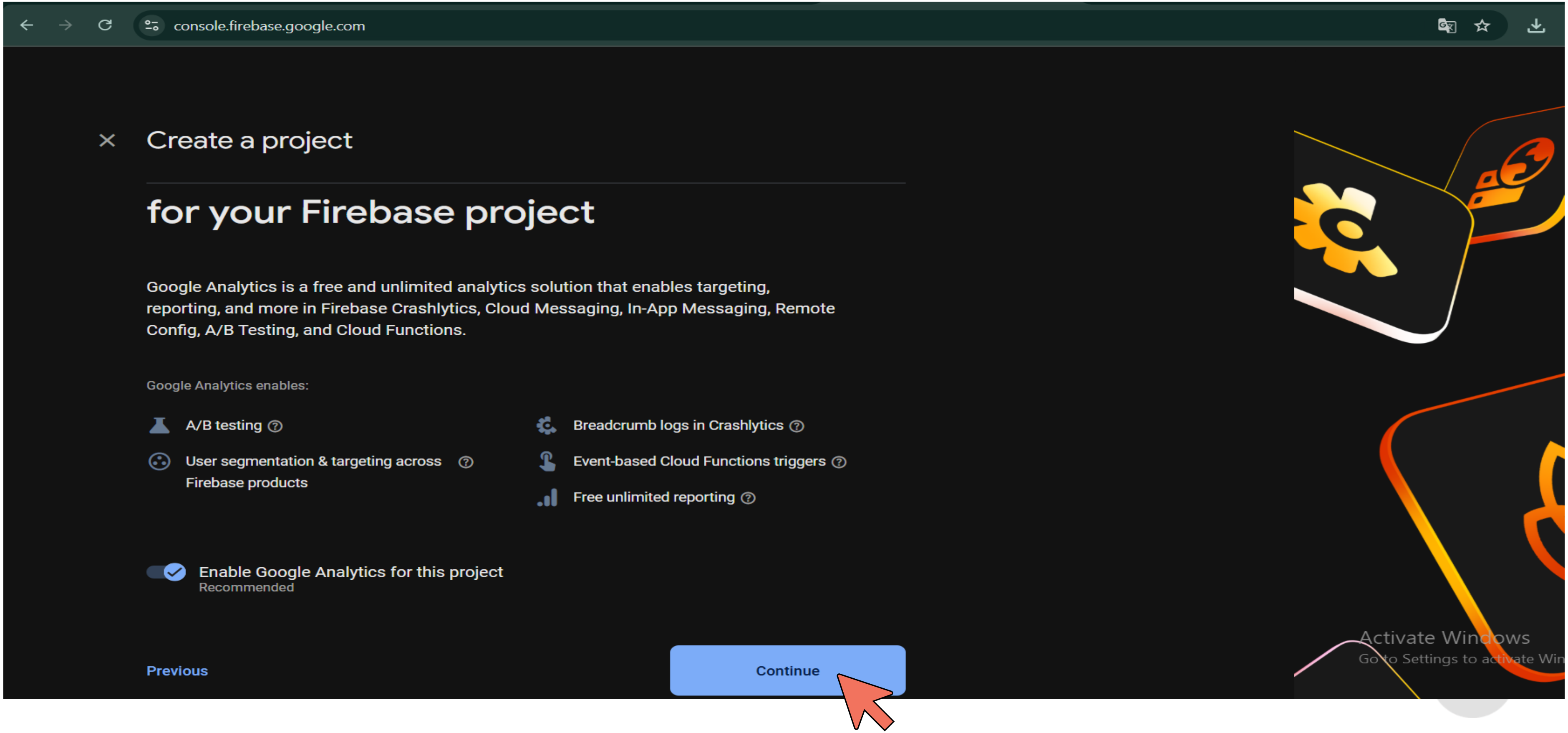
Firebase: Creating a New Project

❖ Write your project name and then continue.



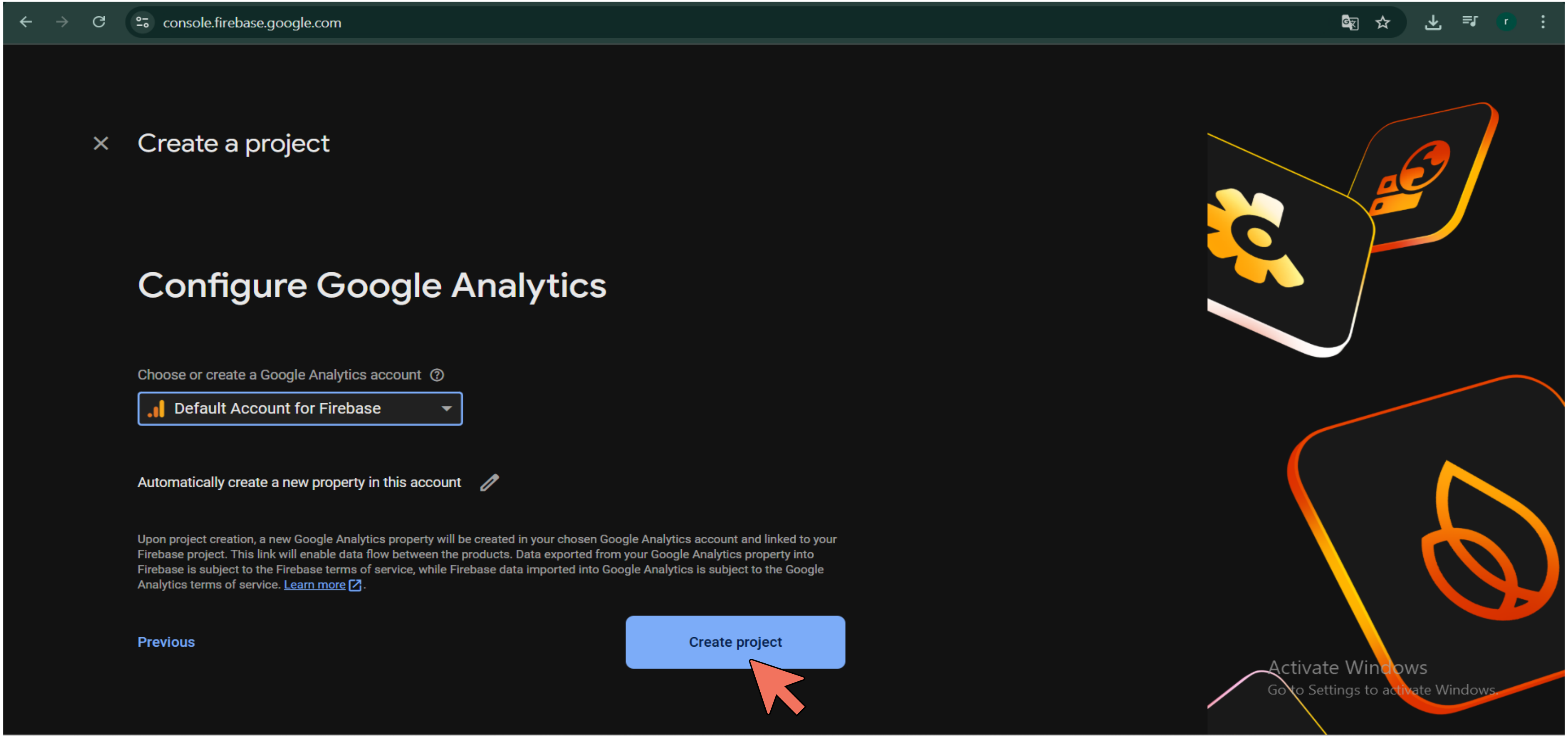
Firebase: Creating a New Project

- Click continue.



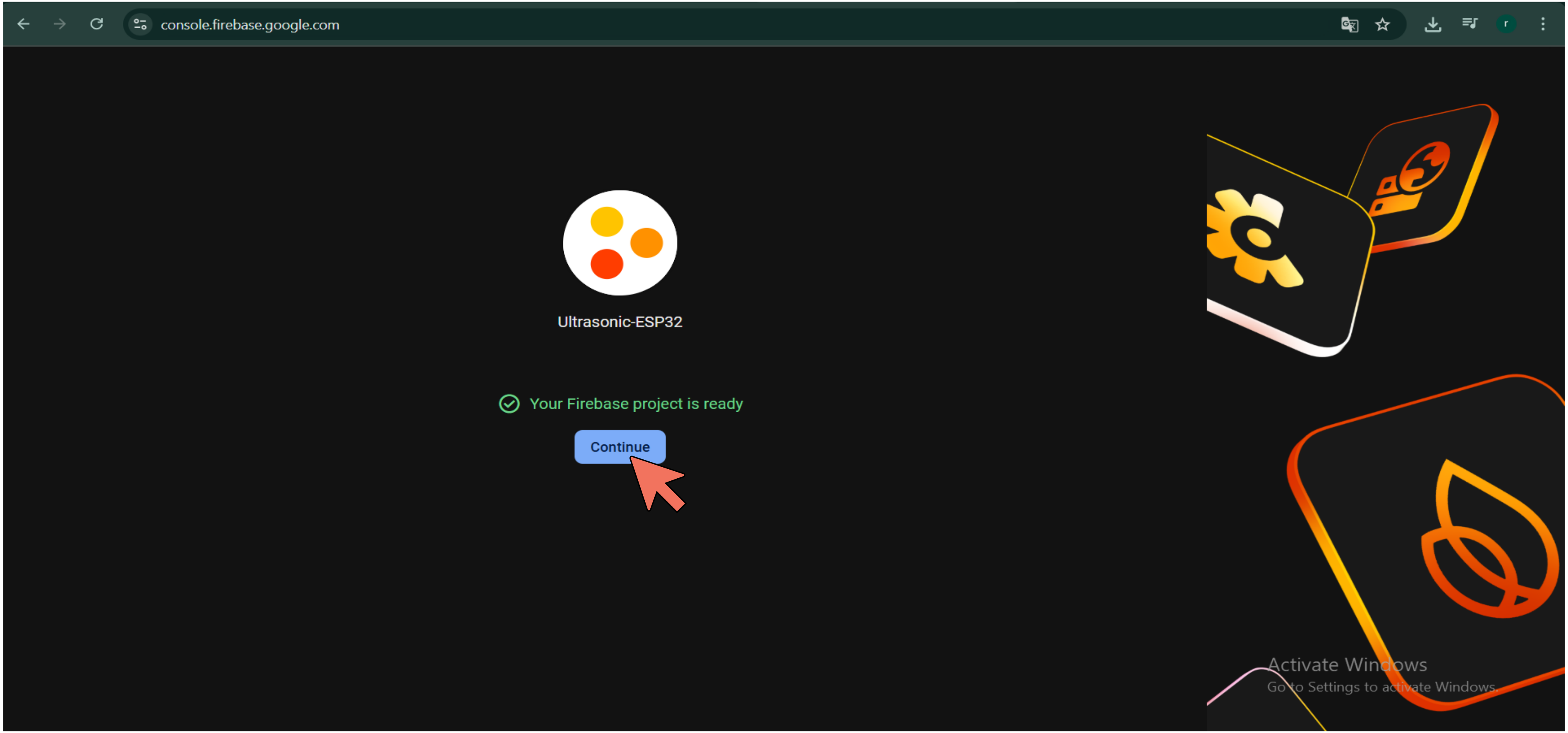
Firebase: Creating a New Project

❖ Choose Default Account for firebase and then click on create project.



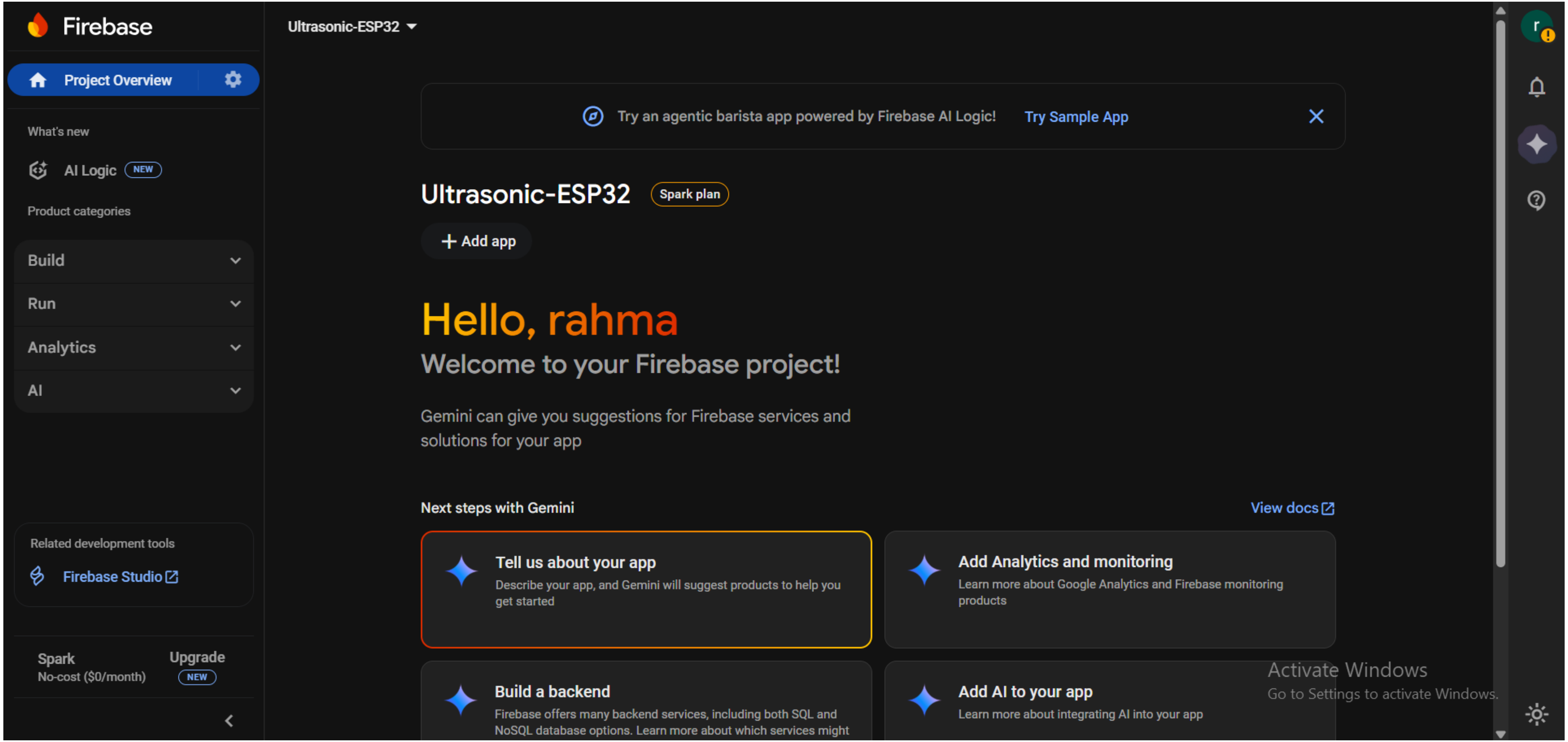
Firebase: Creating a New Project

❖ Wait until project is created ,then click continue.



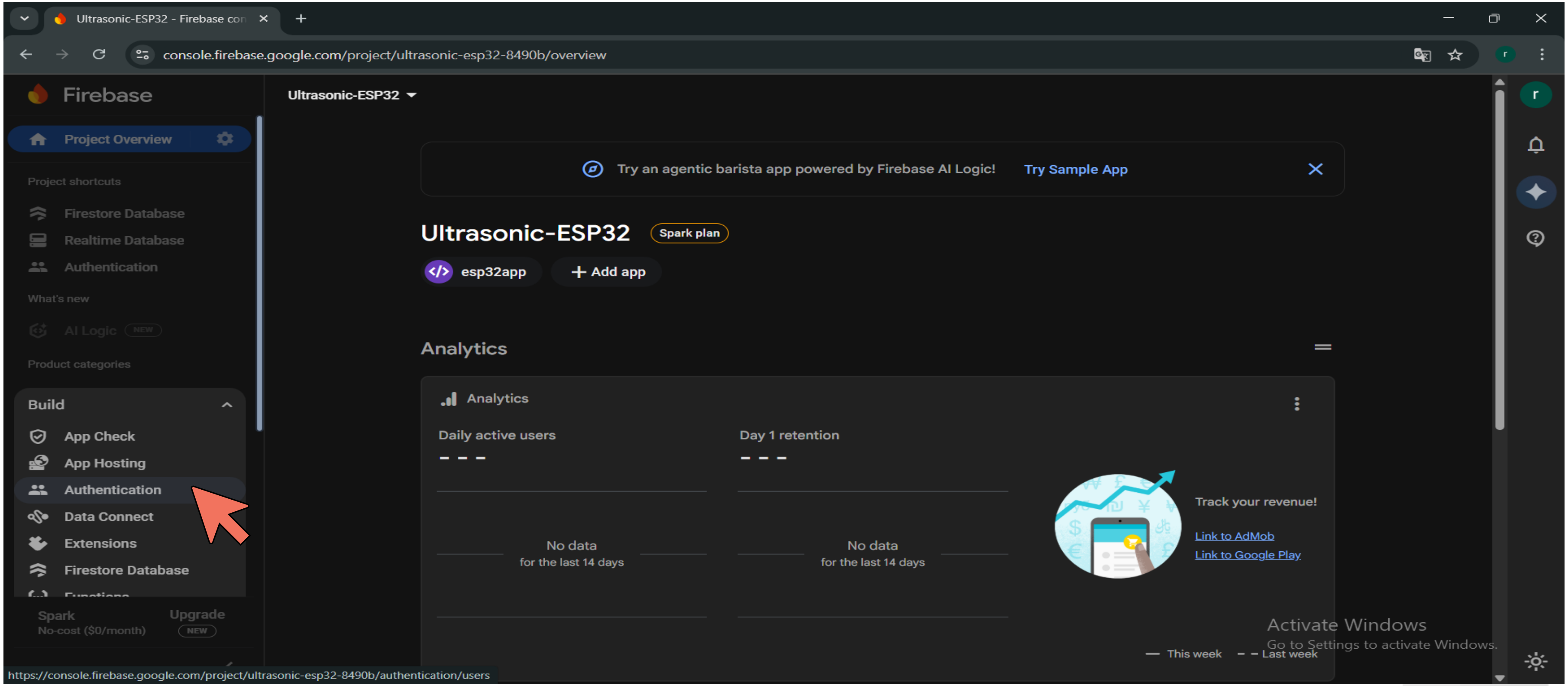
Firebase: Creating a New Project

- You will be redirected to your project consol page.



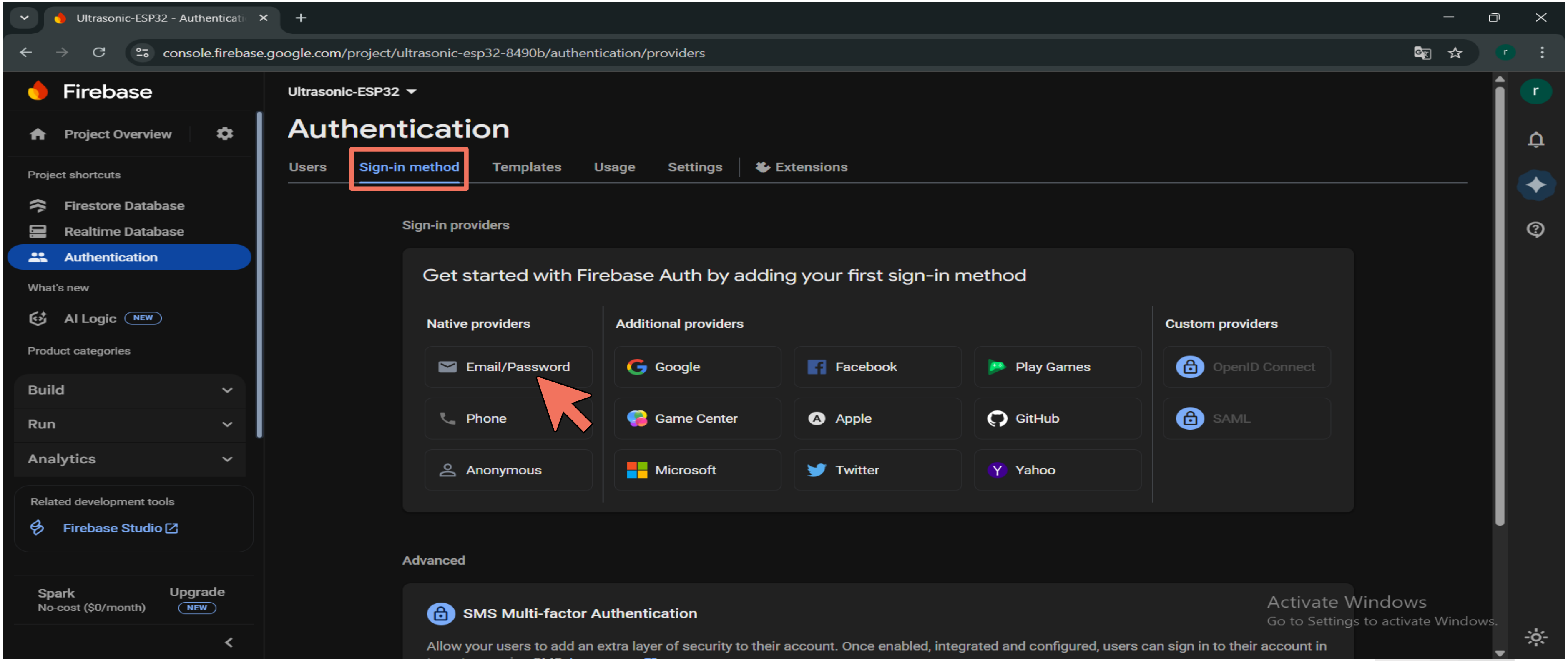
Firebase: Creating a New Project

❖ Expand Build tap and go to Authentication.



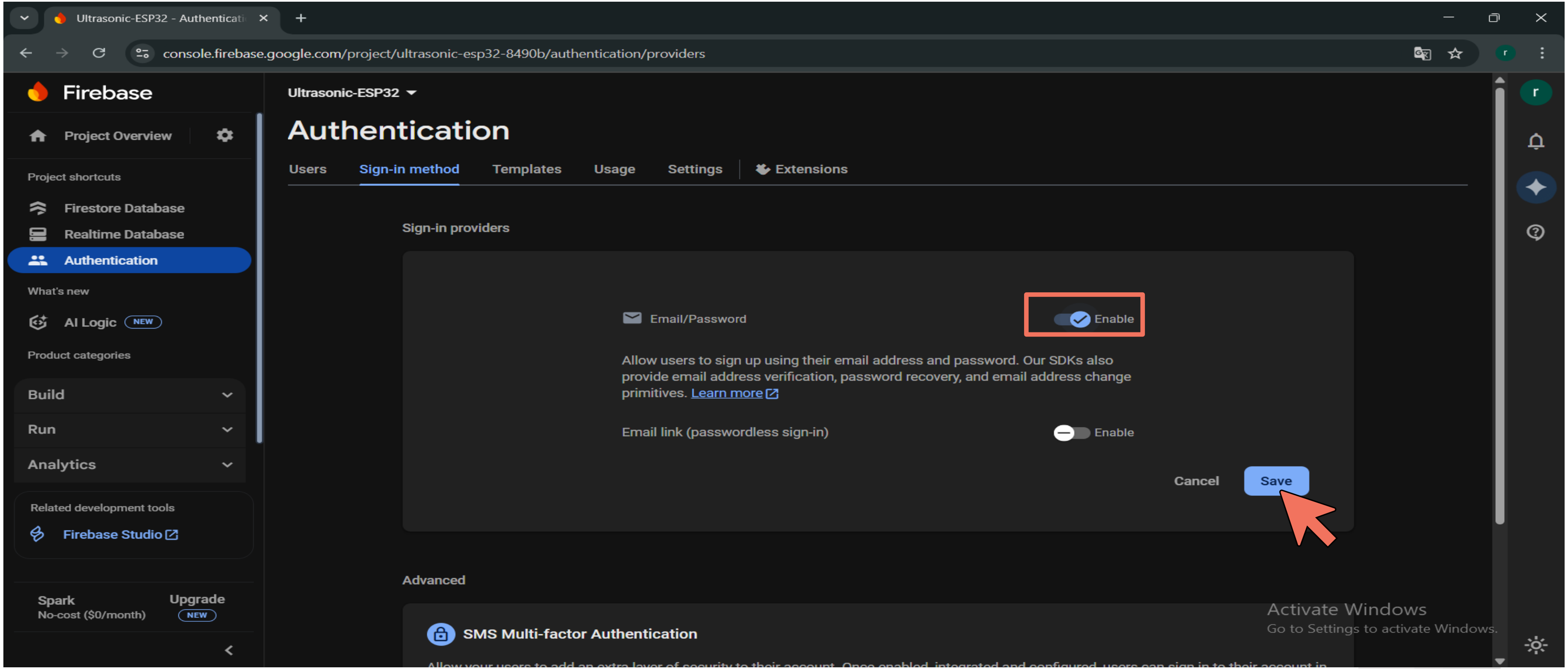
Firebase: Creating a New Project

❖ Go to Sign-in method tap and click on Email/password.



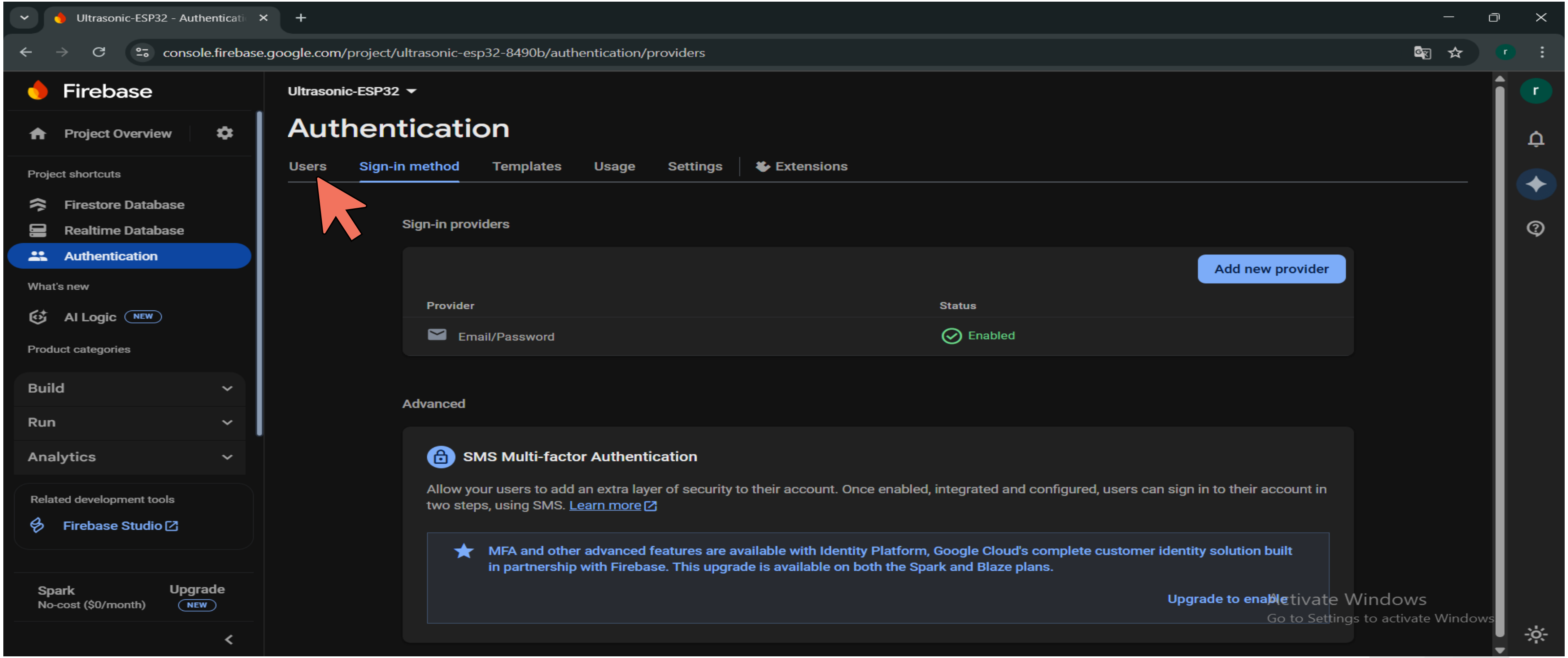
Firebase: Creating a New Project

❖ Click on Enable then click on save.



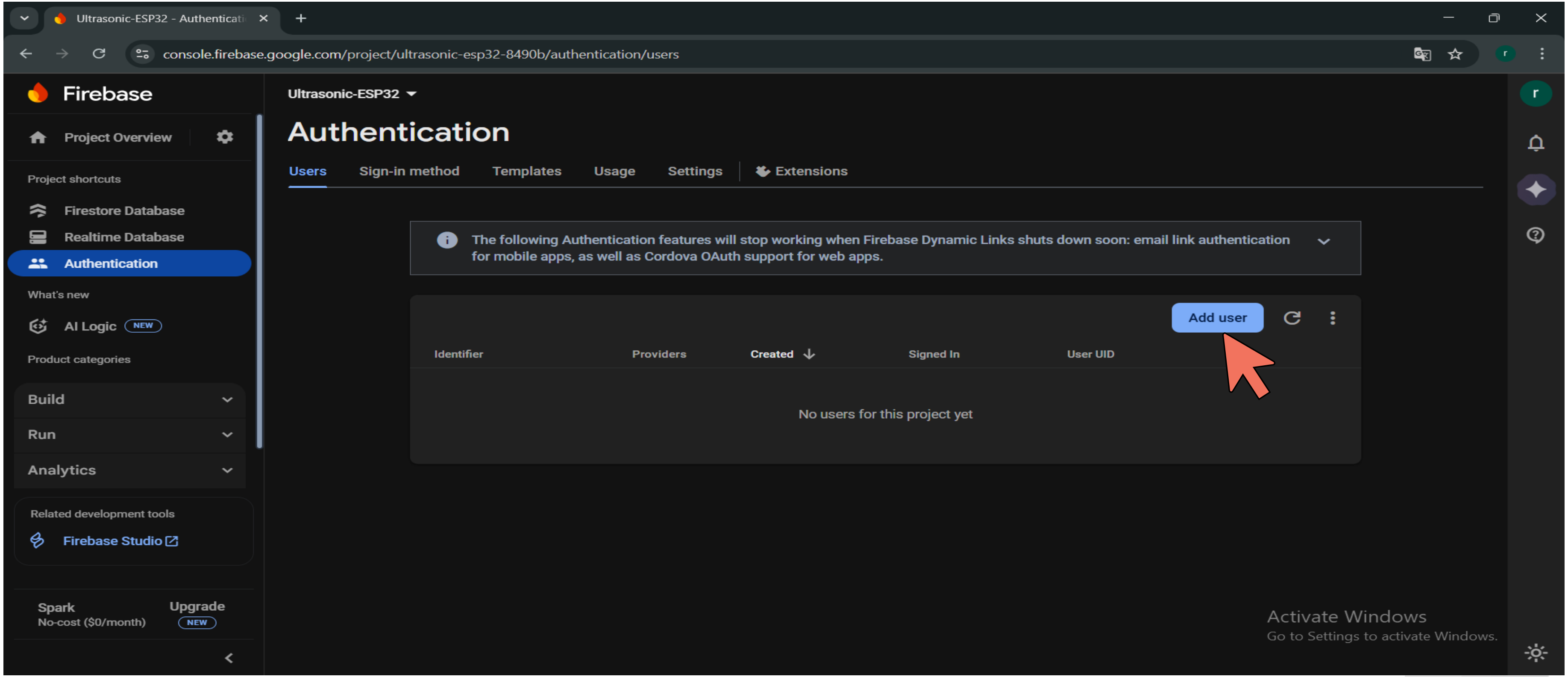
Firebase: Creating a New Project

❖ Now you have created a provider, after that click on Users.



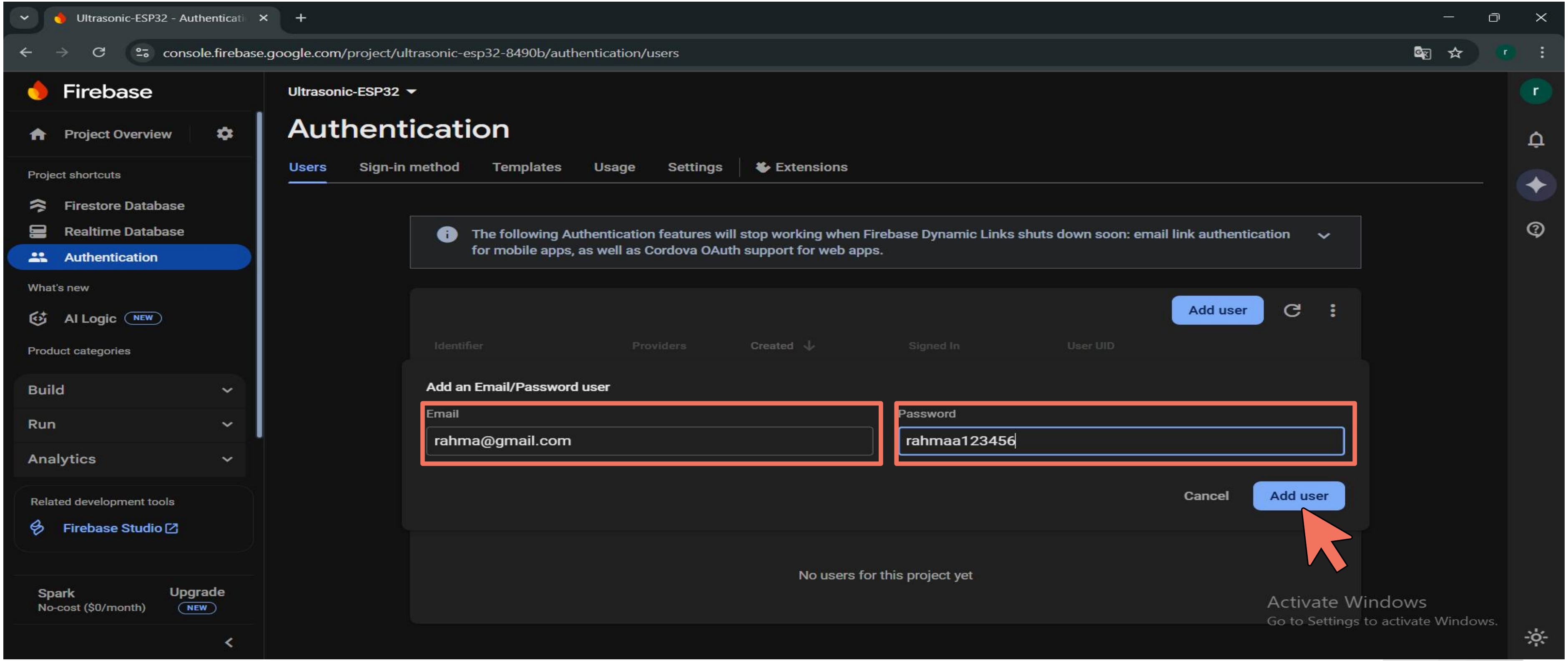
Firebase: Creating a New Project

- click on Add User.



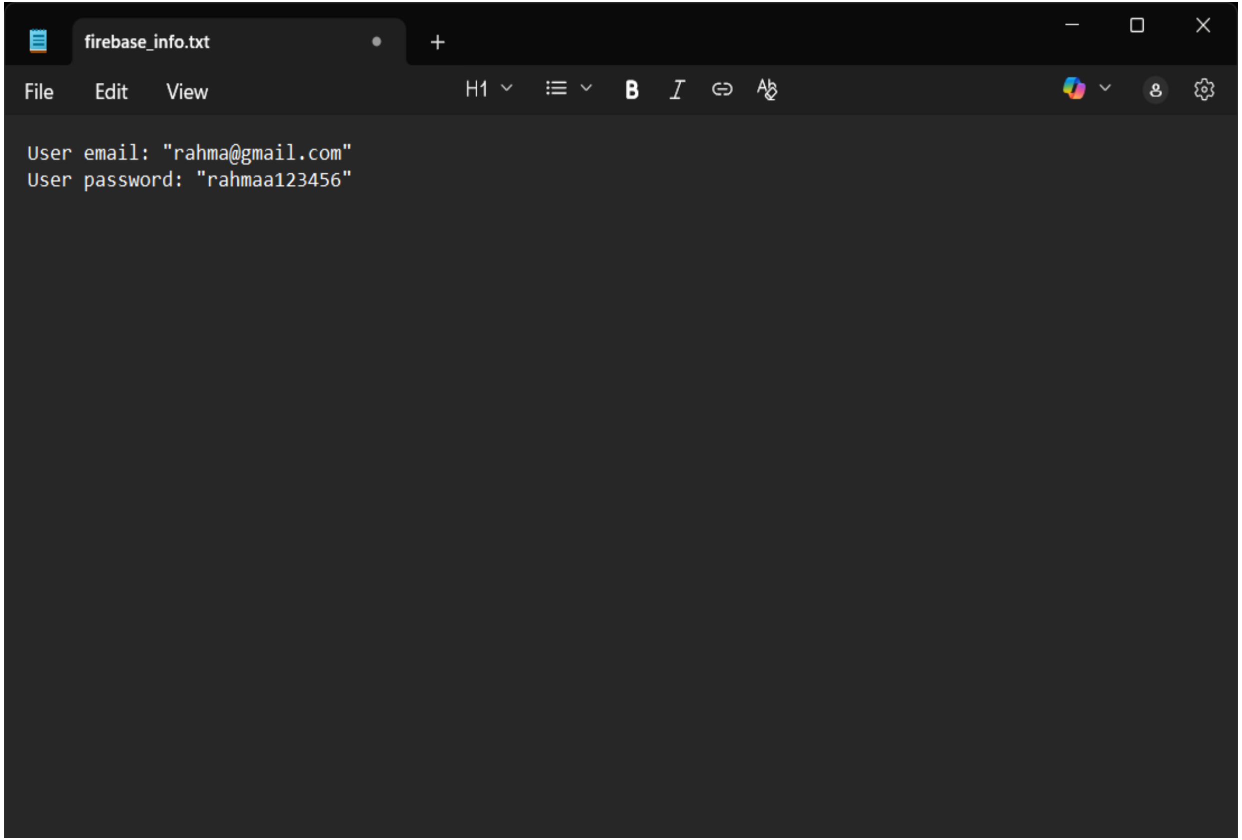
Firebase: Creating a New Project

- Write a user email and password then click on Add User.



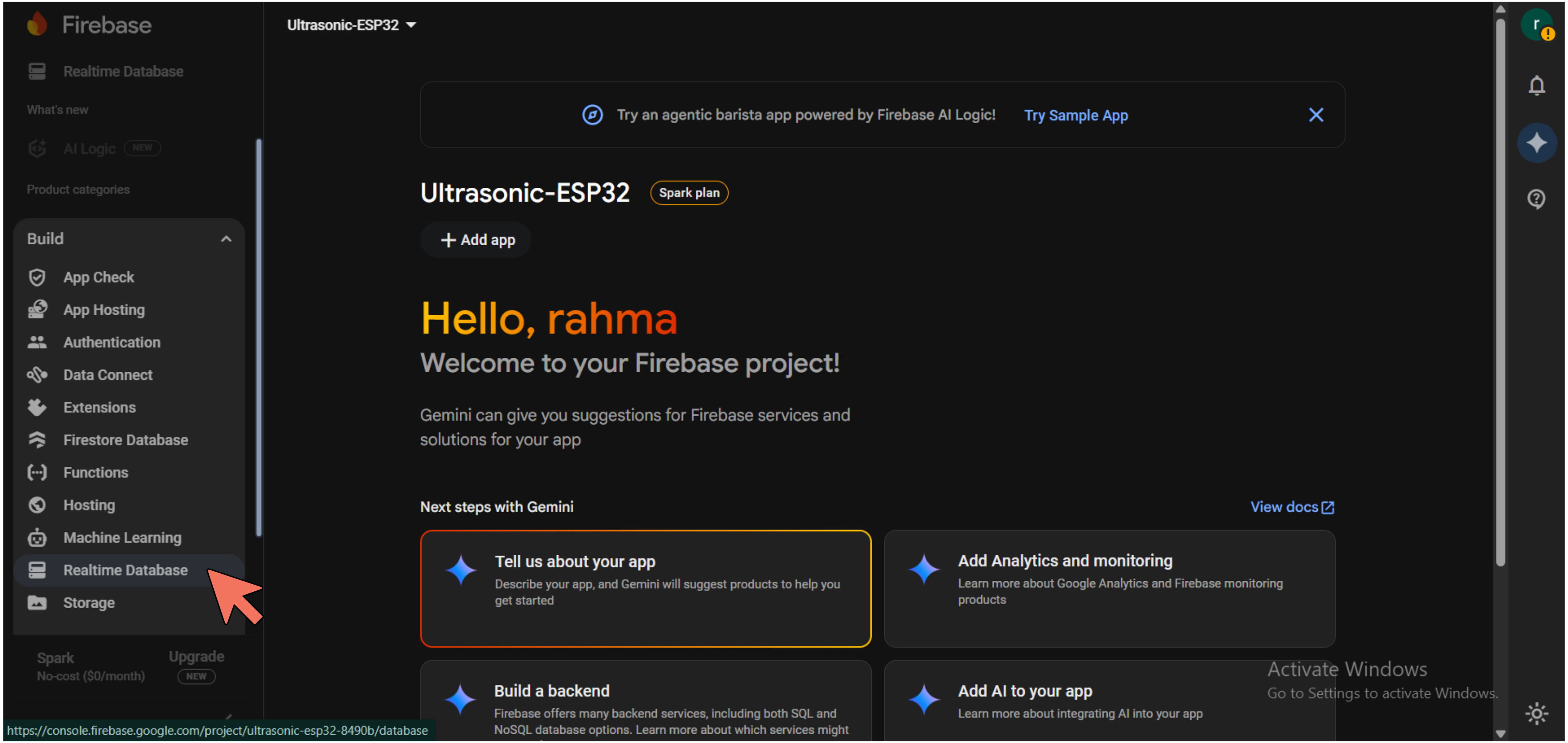
Firebase: Creating a Realtime Database

❖ Save your user email and password in file to use them later in the ESP32 code.



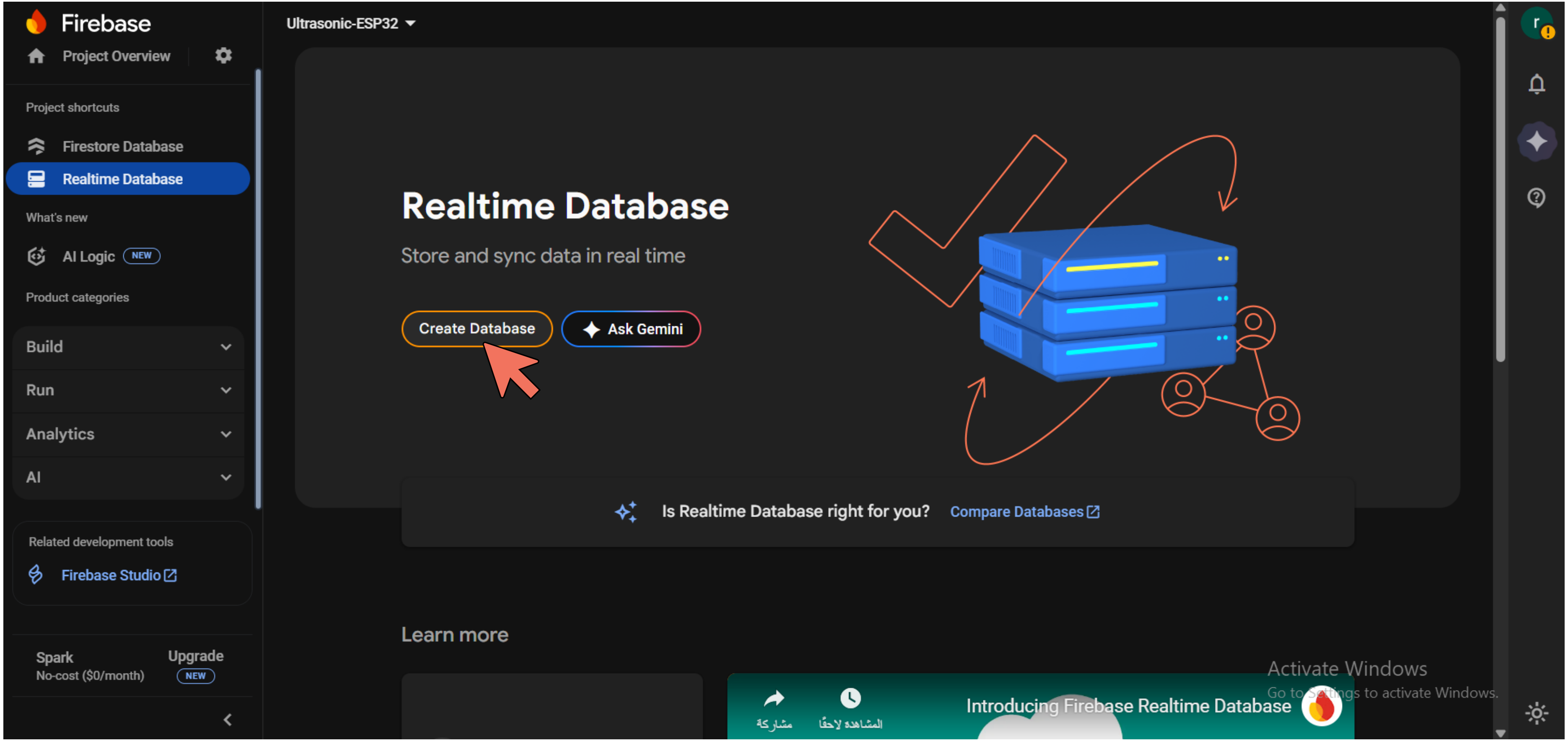
Firebase: Creating a Realtime Database

❖ Then, Expand Build tap and go to Realtime Database.



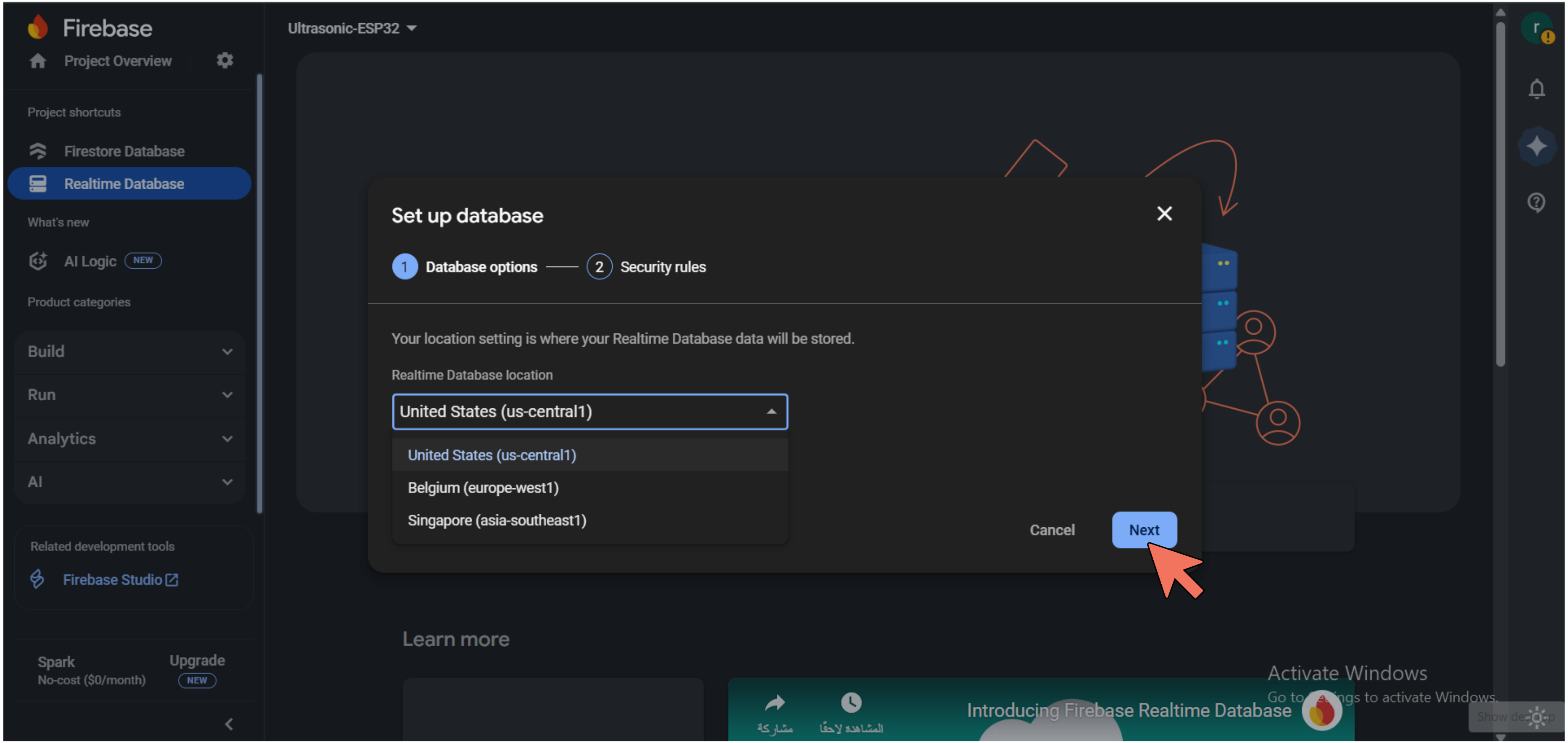
Firebase: Creating a Realtime Database

❖ Click on Create Database.



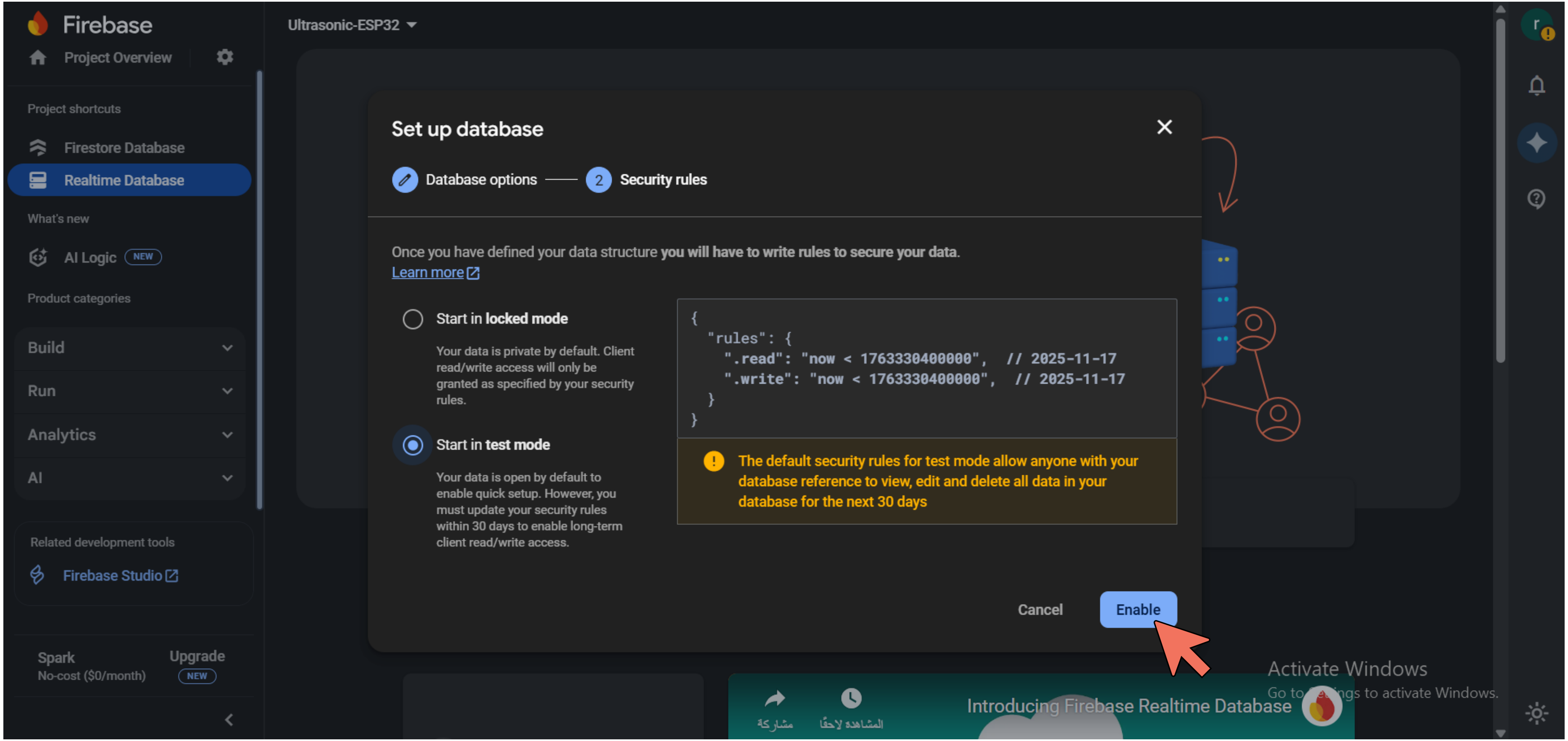
Firebase: Creating a Realtime Database

- Select United States as your database location then and click Next.



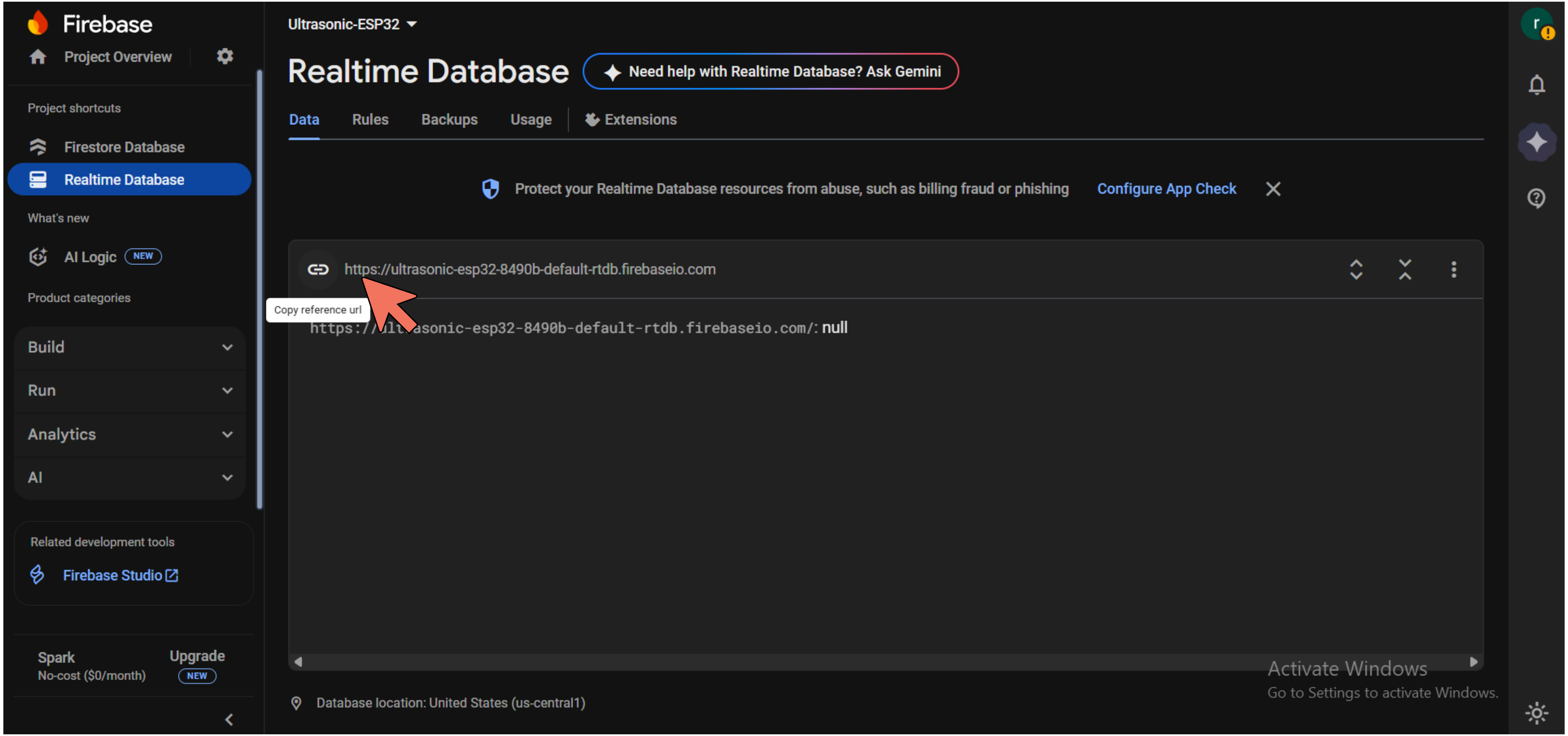
Firebase: Creating a Realtime Database

❖ Select Start in test mode then click enable.I chose *start in test mode* to easily read and write data during testing without authentication



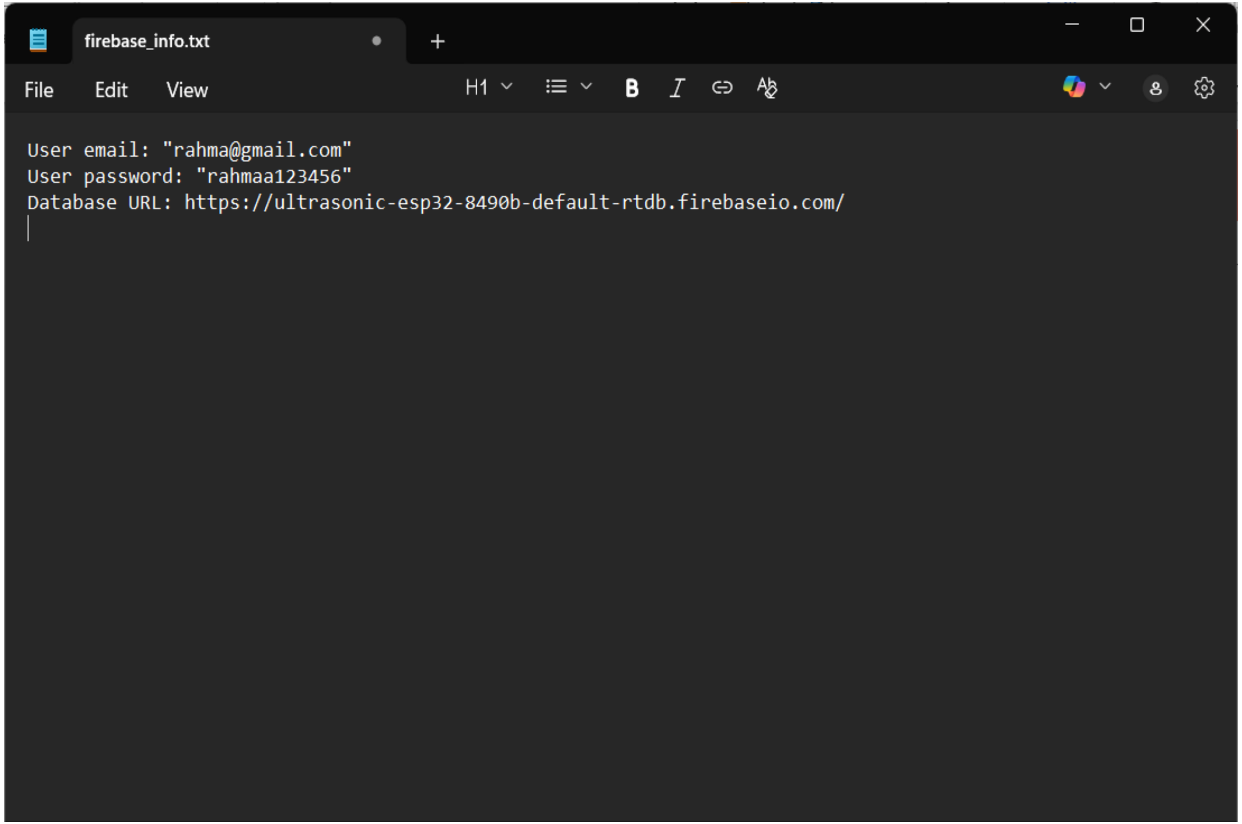
Firebase: Creating a Realtime Database

❖ Now your Database is created, you need to copy your database URL to use it in the ESP32 code for connecting to firebase.



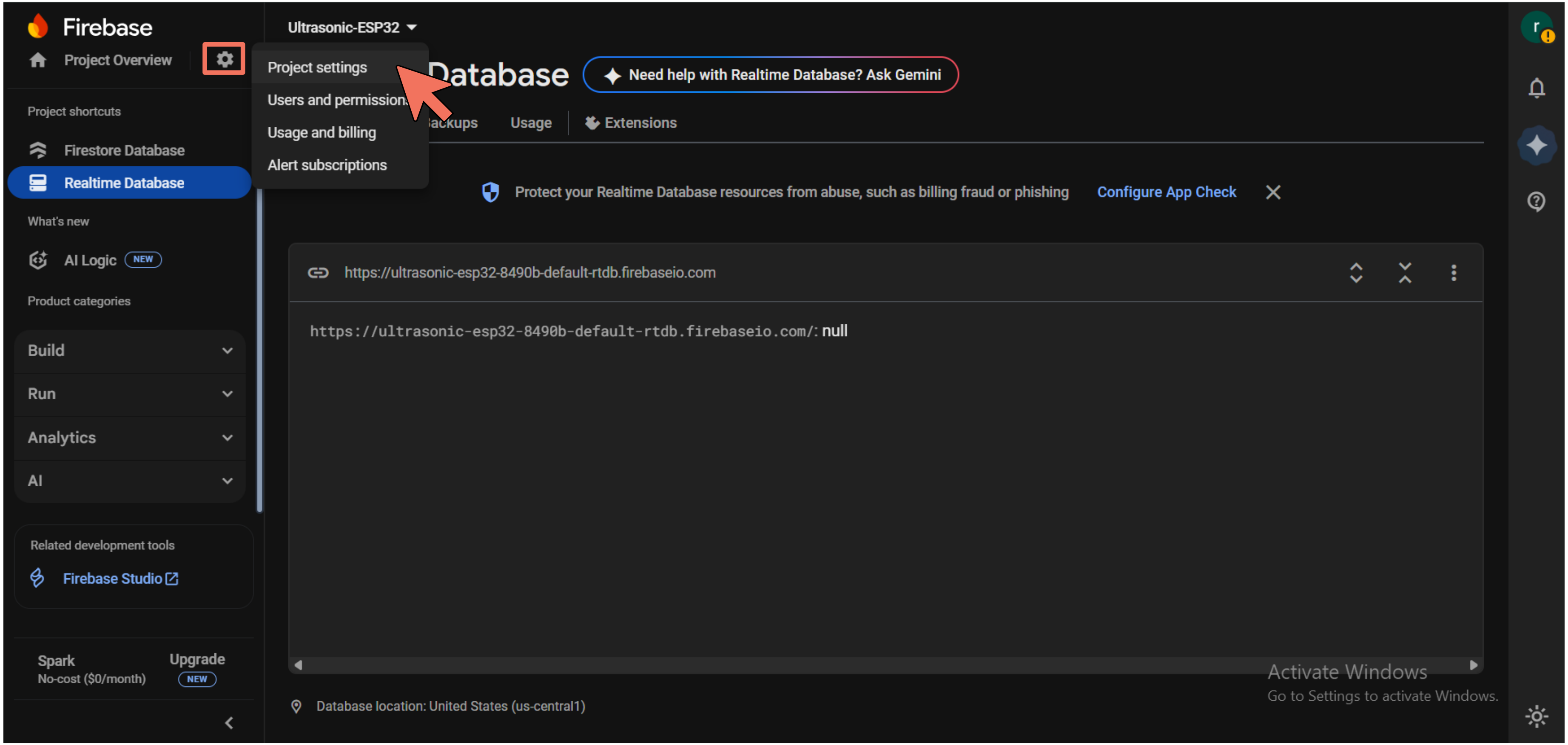
Firebase: Creating a Realtime Database

❖ Update your Notepad file with your Database URL to use them later in the ESP32 code.



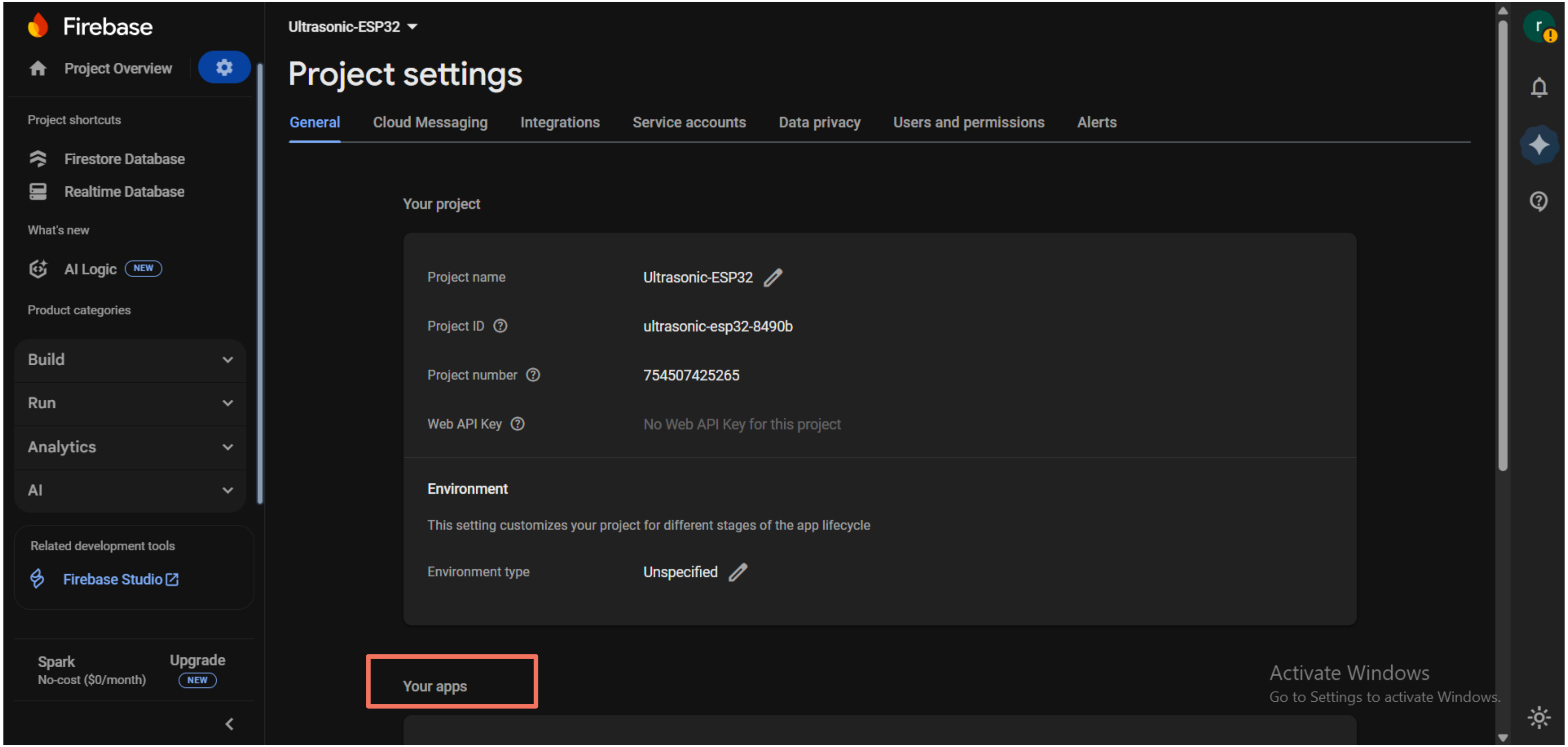
Firebase: Creating a Realtime Database

❖Go to project settings.



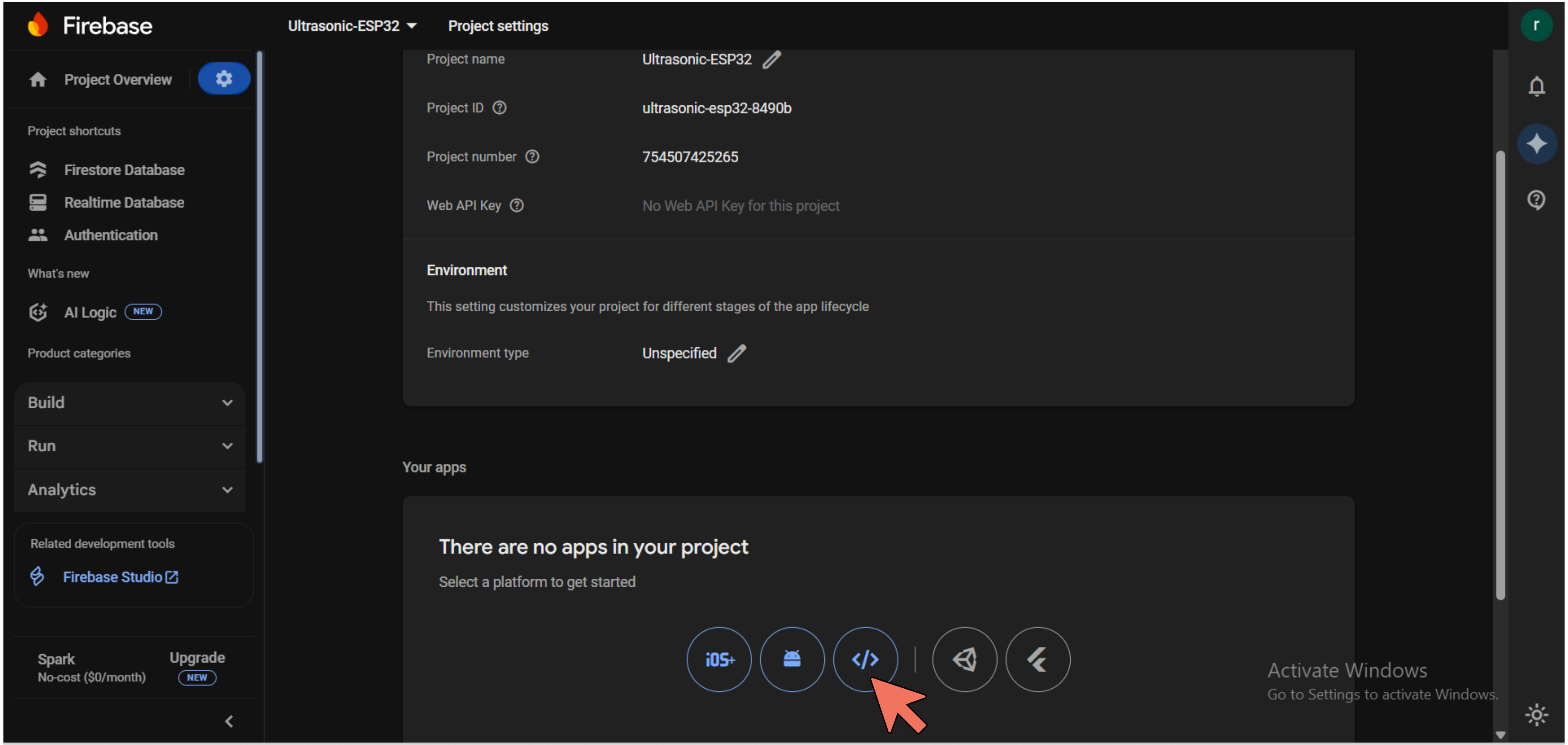
Firebase: Creating a Realtime Database

❖Scroll down until you reach your apps window.



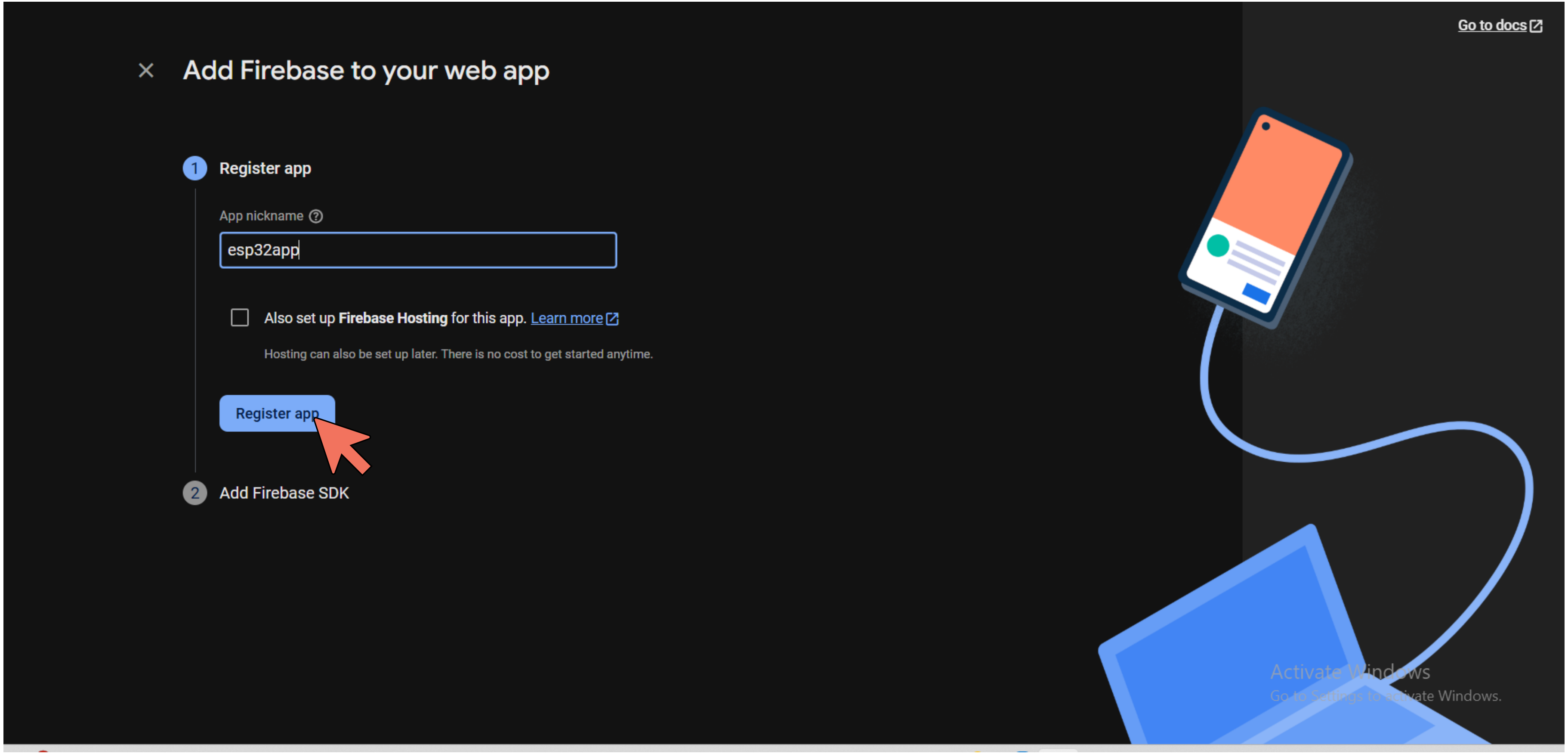
Firebase: Creating a Realtime Database

❖Select web app icon .



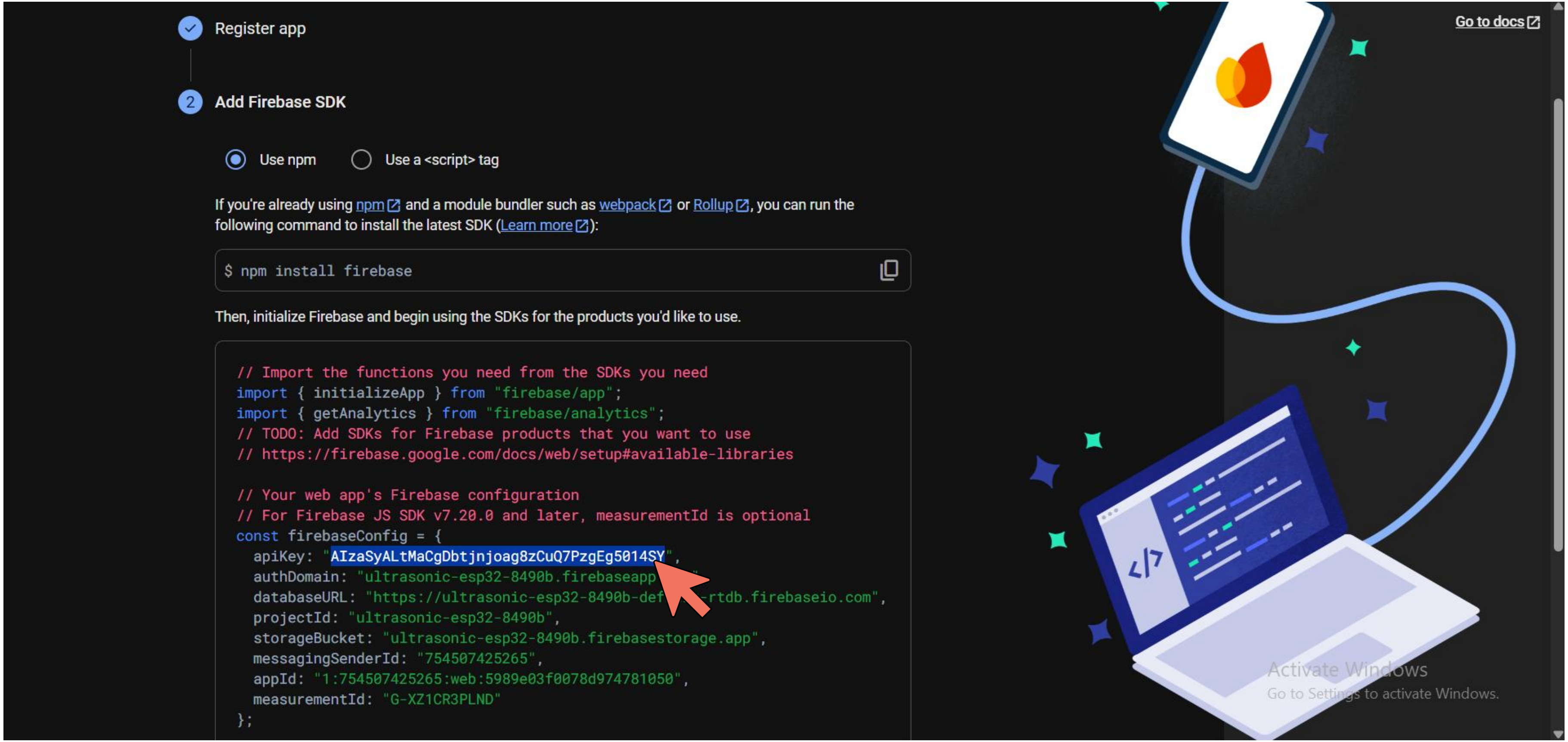
Firebase: Creating a Realtime Database

❖Write your App name and then click Register app.



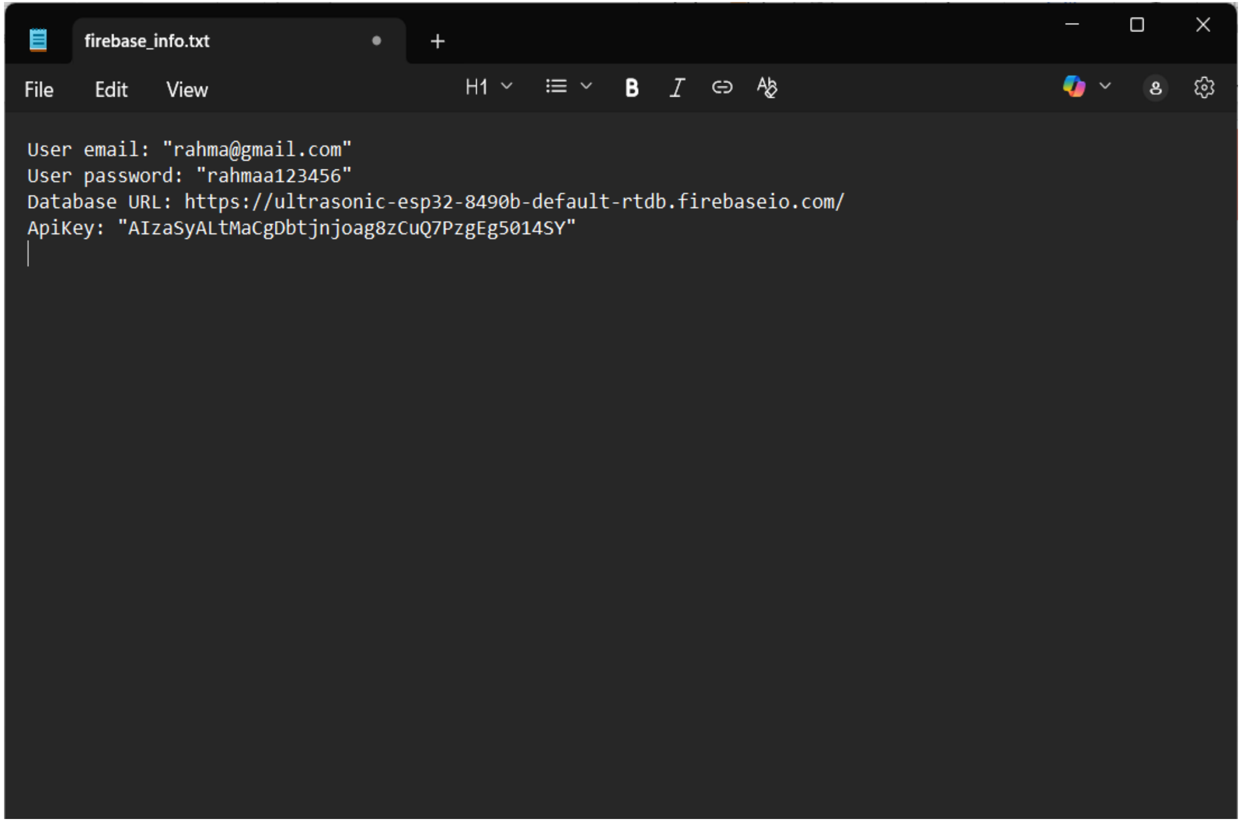
Firebase: Creating a Realtime Database

❖ Copy this ApiKey.



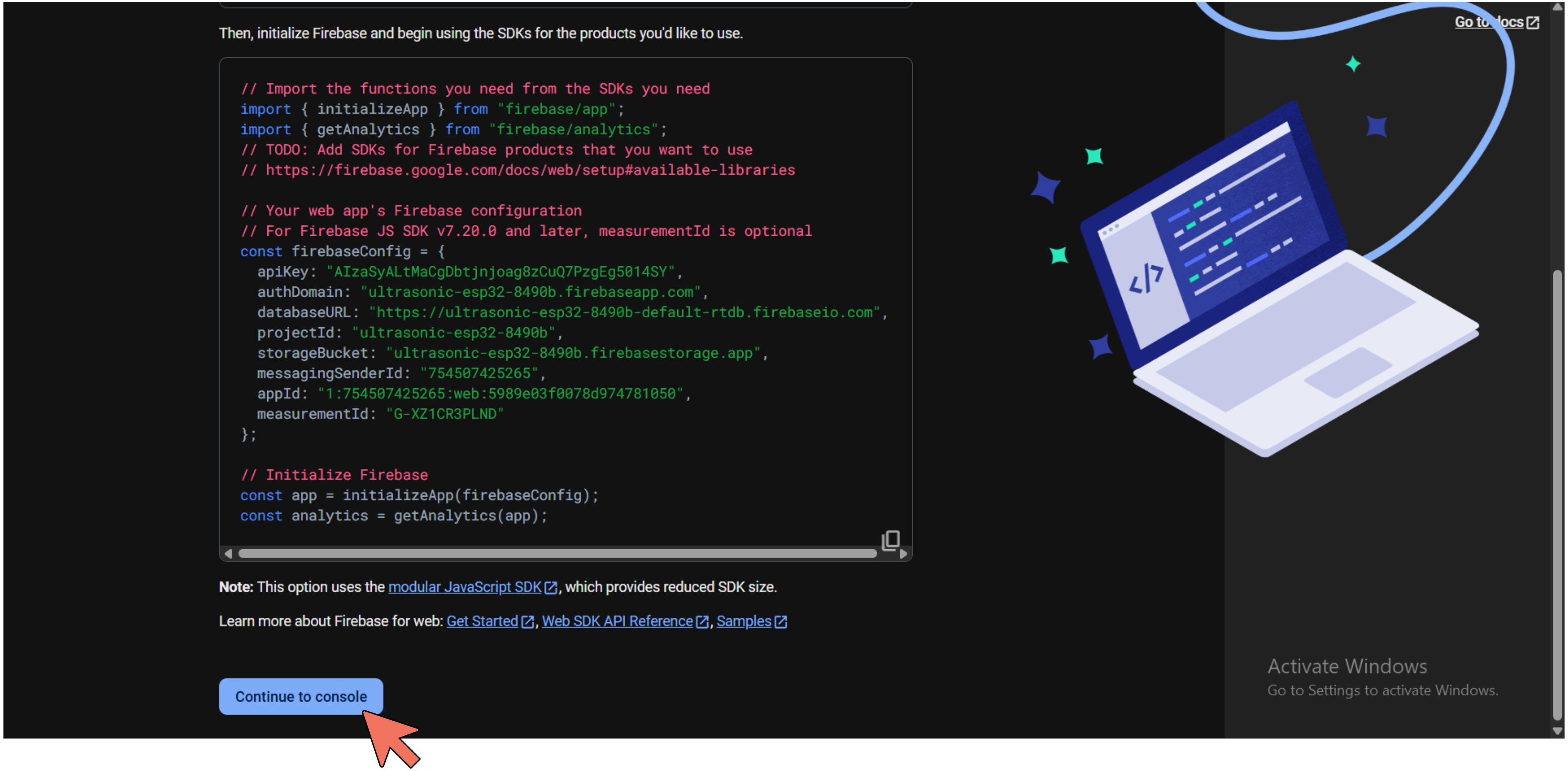
Firebase: Creating a Realtime Database

❖ Update your Notepad file with your ApiKey to use them later in the ESP32 code.



Firebase: Creating a Realtime Database

❖Scroll down and click continue to consol.



Firebase with IoT

- All Firebase Realtime Database data is stored as JSON objects.
- You can think of the database as a cloud-hosted JSON tree.
- Unlike a SQL database, there are no tables or records.
- When you add data to the JSON tree, it becomes a node in the existing JSON structure with an associated key.

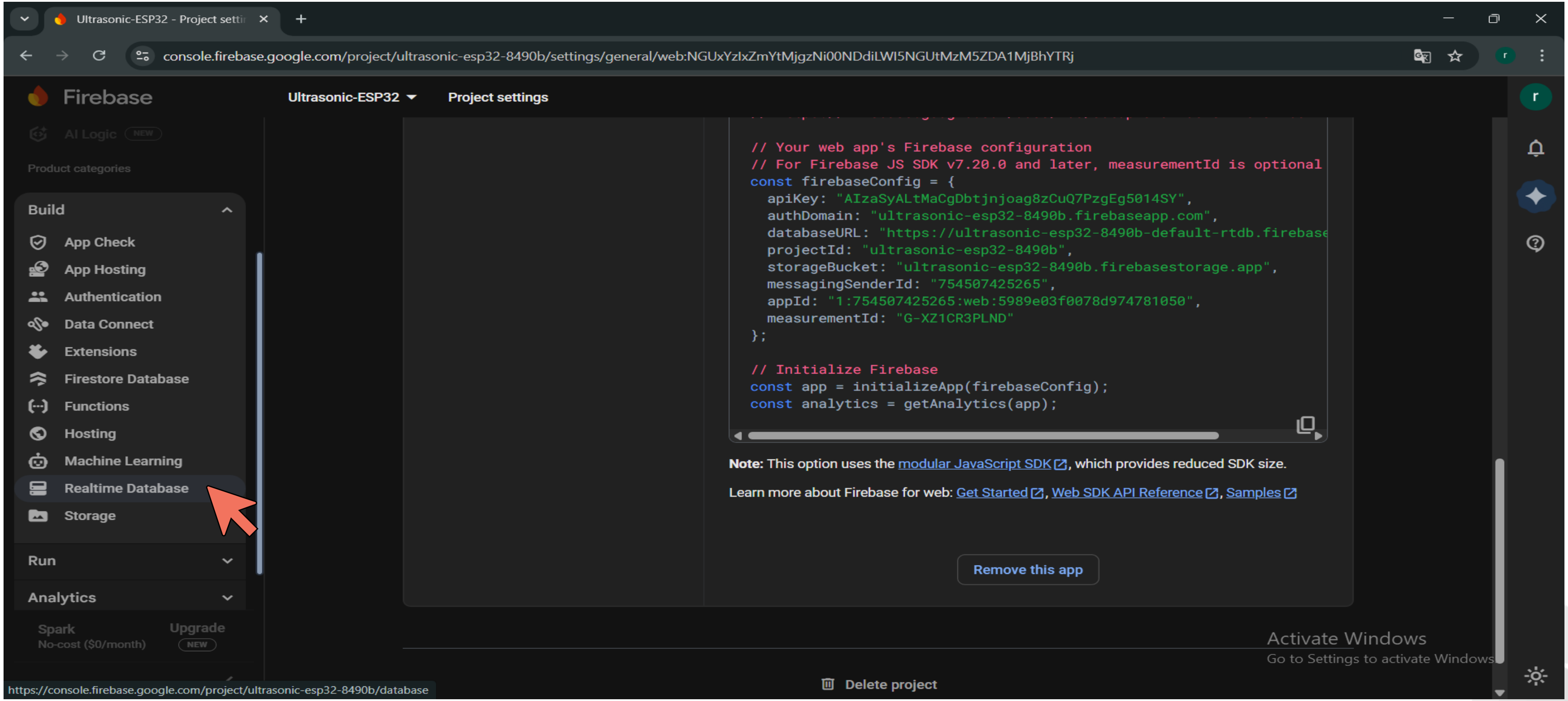
```
{
    "smartHome": {
        "kitchen": {
         "light": "off",
        "temperature": 20
        },
        "bedroom": {
            "light": "on",
            "temperature"
            : 18
        }
    }
}
```

Firebase with IoT


```json
{
  "smartHome": {
    "devices": {
      "light": {
        "status": "on",
        "brightness": 75
      },
      "thermostat": {
        "temperature": 22,
        "mode": "auto"
      },
      "door": {
        "status": "closed"
      }
    },
    "security": {
      "alarm": {
        "status": "armed"
      },
      "camera": {
        "status": "active"
      }
    }
  }
}
```

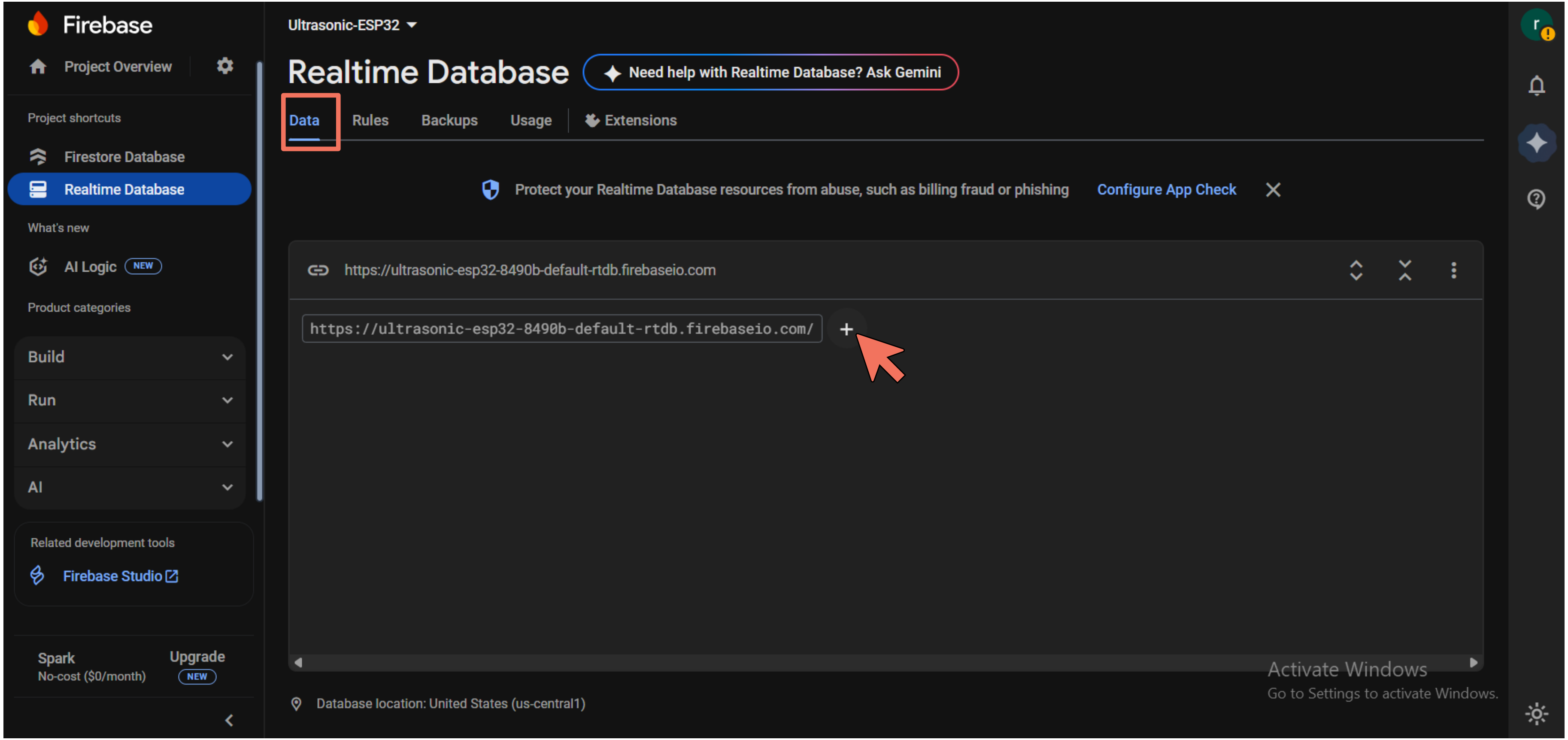
Firebase: Creating a Realtime Database

❖ Return to your Realtime Database.



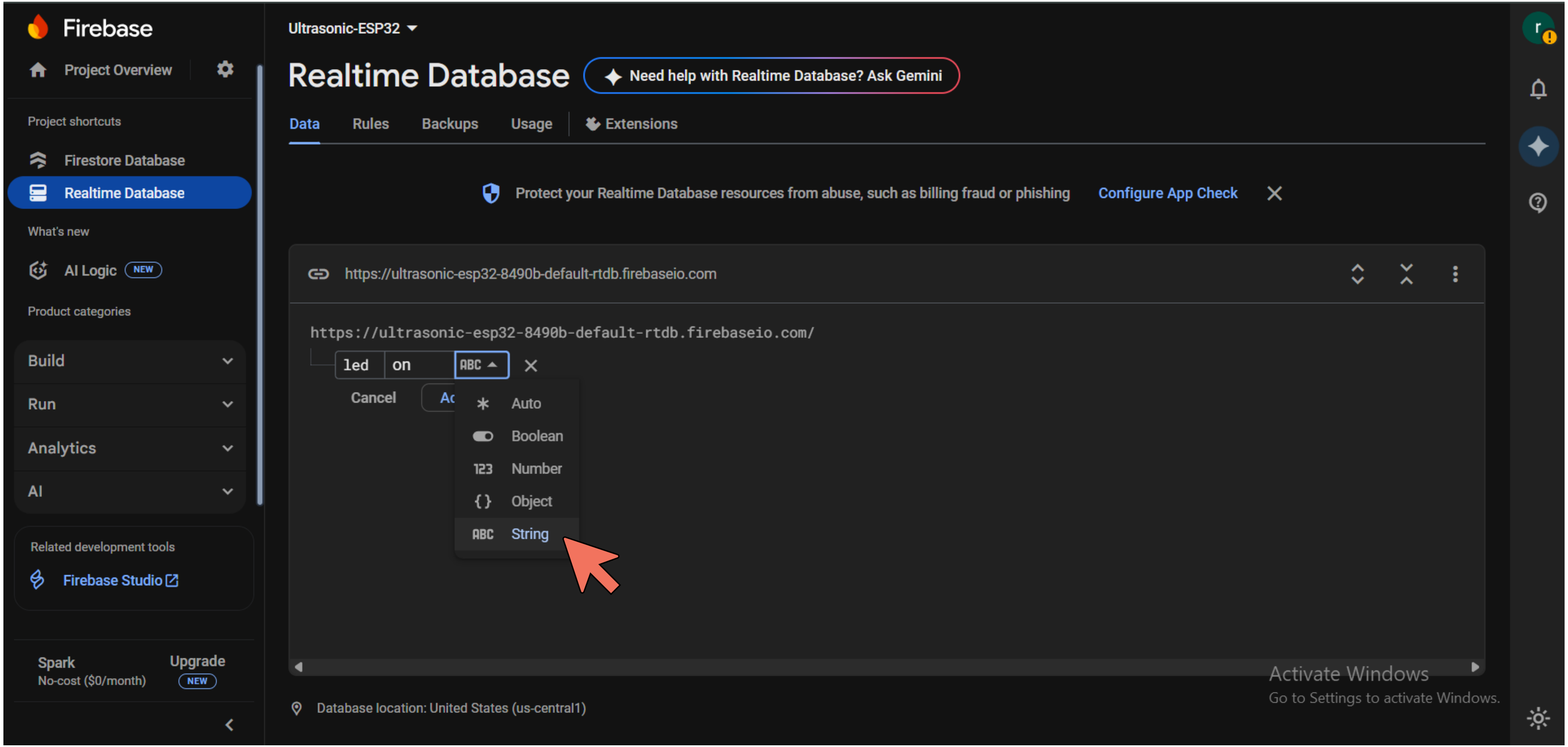
Firebase: Creating a Realtime Database

❖ Select Data tap then Click on add Node.



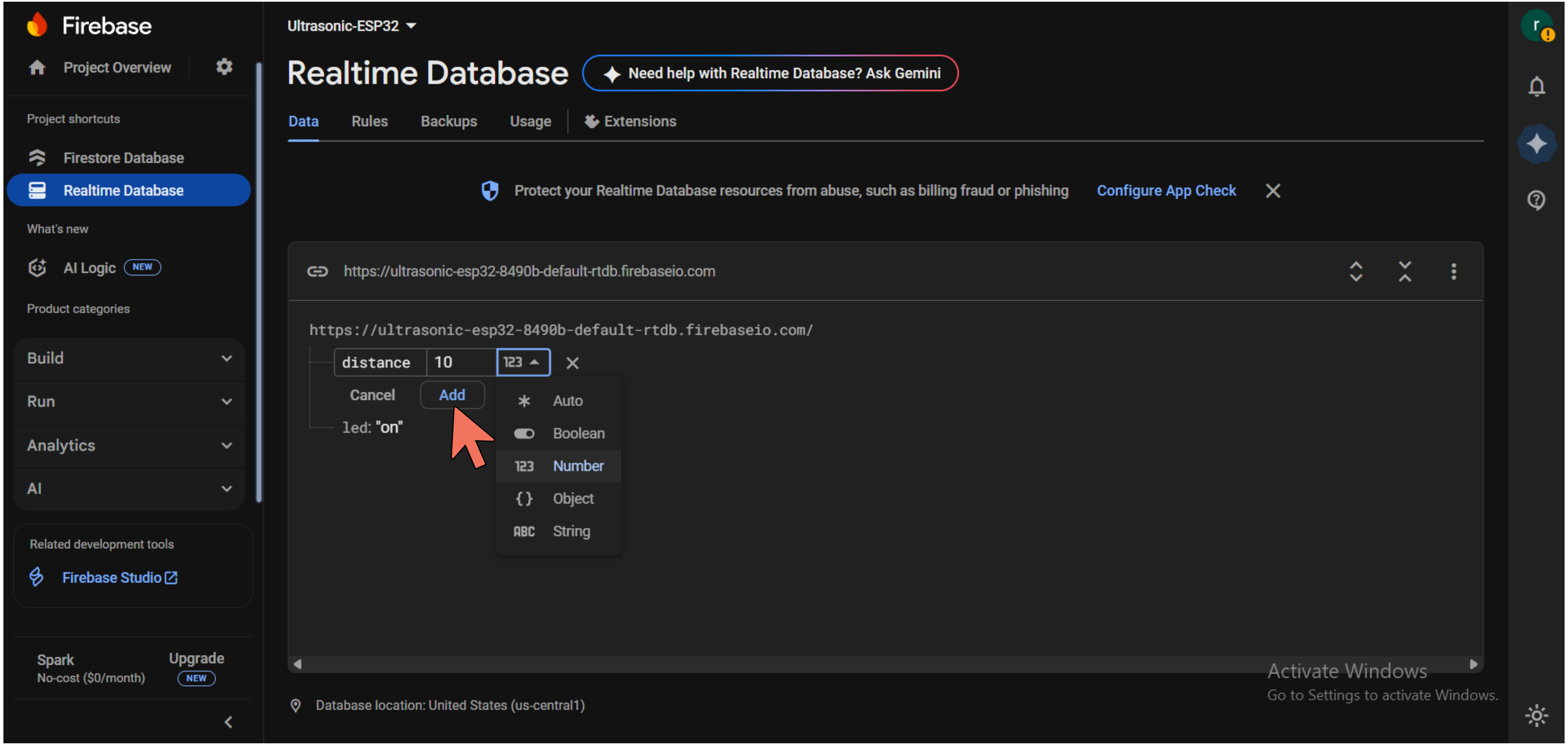
Firebase: Creating a Realtime Database

❖ Write your node Key and value choose its datatype and then click Add.



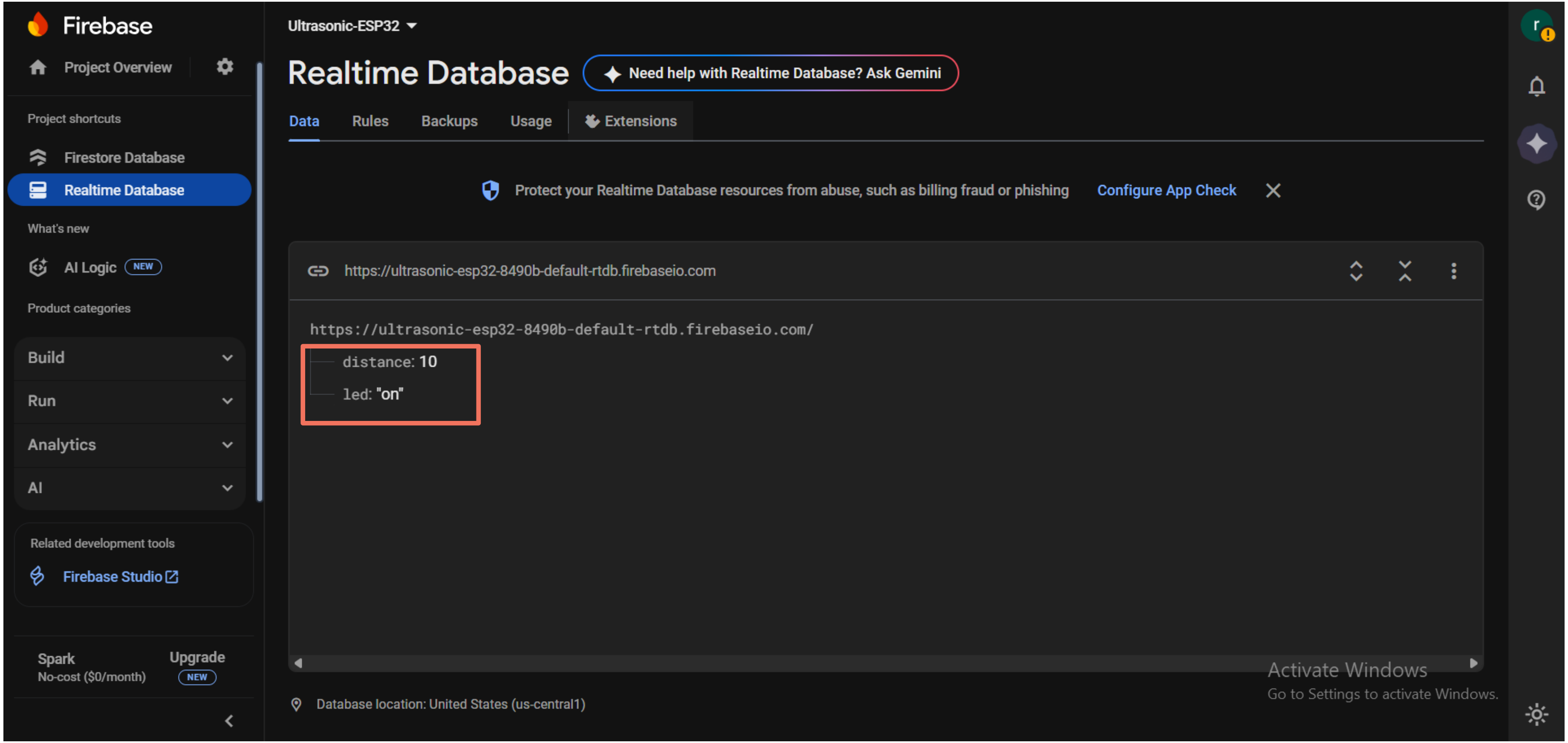
Firebase: Creating a Realtime Database

❖ I added a new node named "distance" and set its value to 10 to represent the ultrasonic sensor reading.



Firebase: Creating a Realtime Database

❖ Now I have two nodes: "led" to control the LED status and "distance" to store the ultrasonic sensor reading.



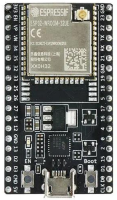
Firebase with IoT

❖ **Now, let's start connecting our Firebase database with our project.**

- I will start building my project, which consists of an **ultrasonic sensor** and an **LED** connected to the **ESP32**.
- The ultrasonic sensor will measure the distance, and the measured value will be sent to **Firebase**.
- Based on the distance value, the LED will turn **ON if distance < 10 or OFF otherwise** automatically.

Project Component we need !

1 ESP-32 Development Board WiFi Bluetooth.



2 breadboard.

3 Ultrasonic Sensor (HC-SR04).

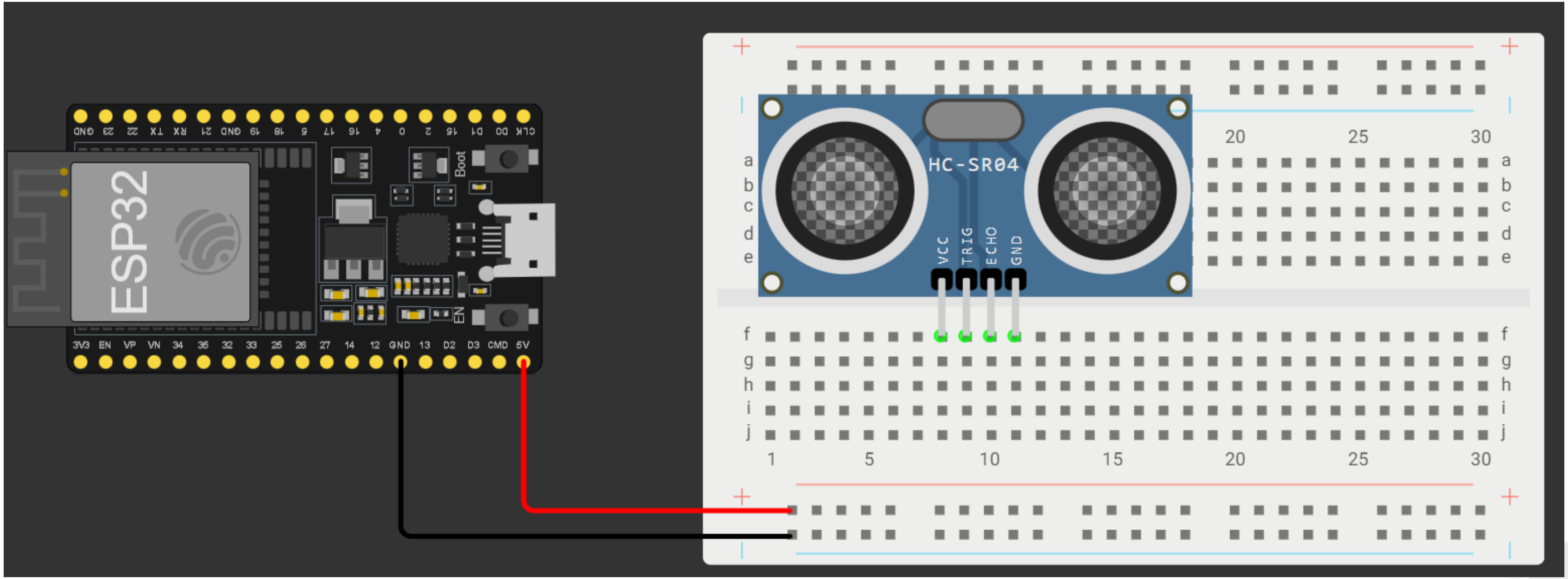
4 Led

5 Resistor 220ohm

6 Wires

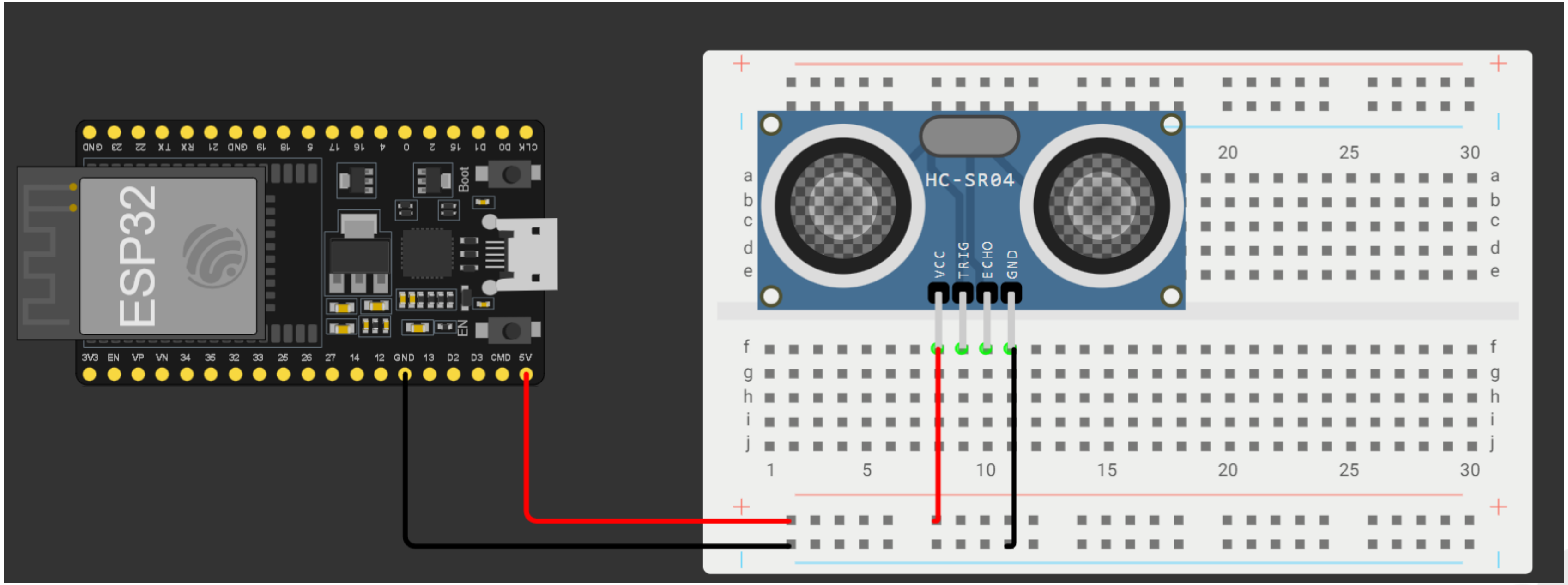
Ultrasonic Sensor with Led

Step1:Connecting the Ultrasonic Sensor to the Breadboard



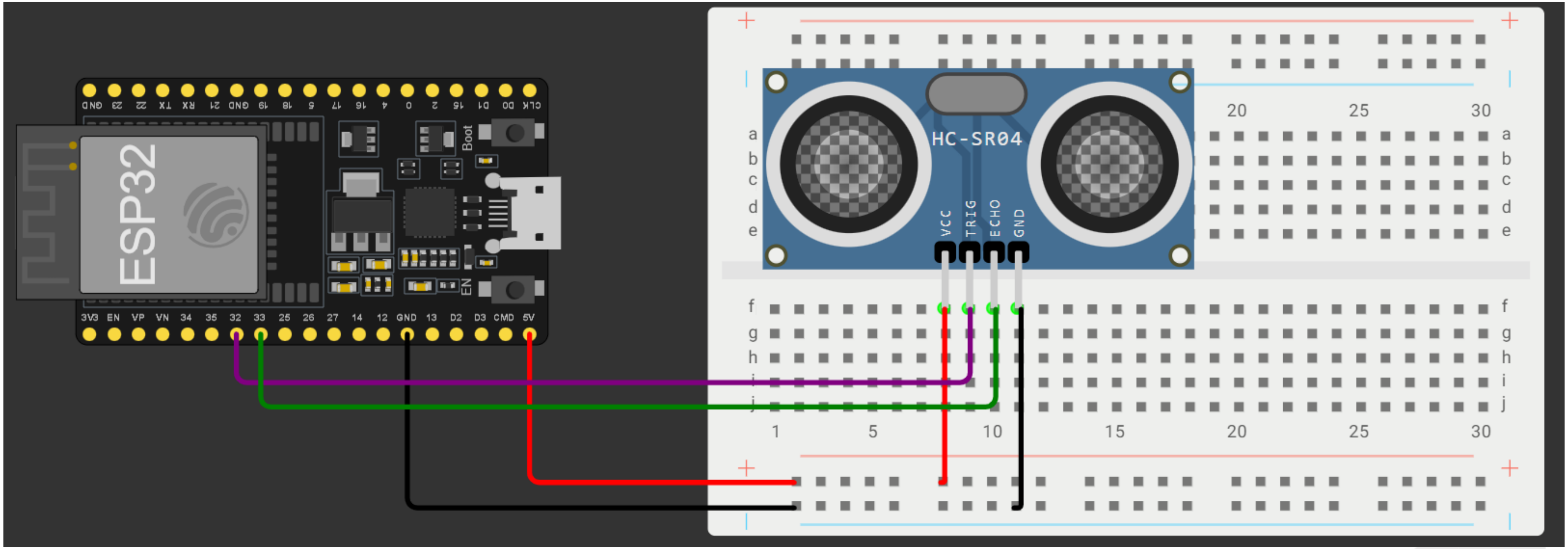
Ultrasonic Sensor with Led

Step 2: Powering the Ultrasonic Sensor



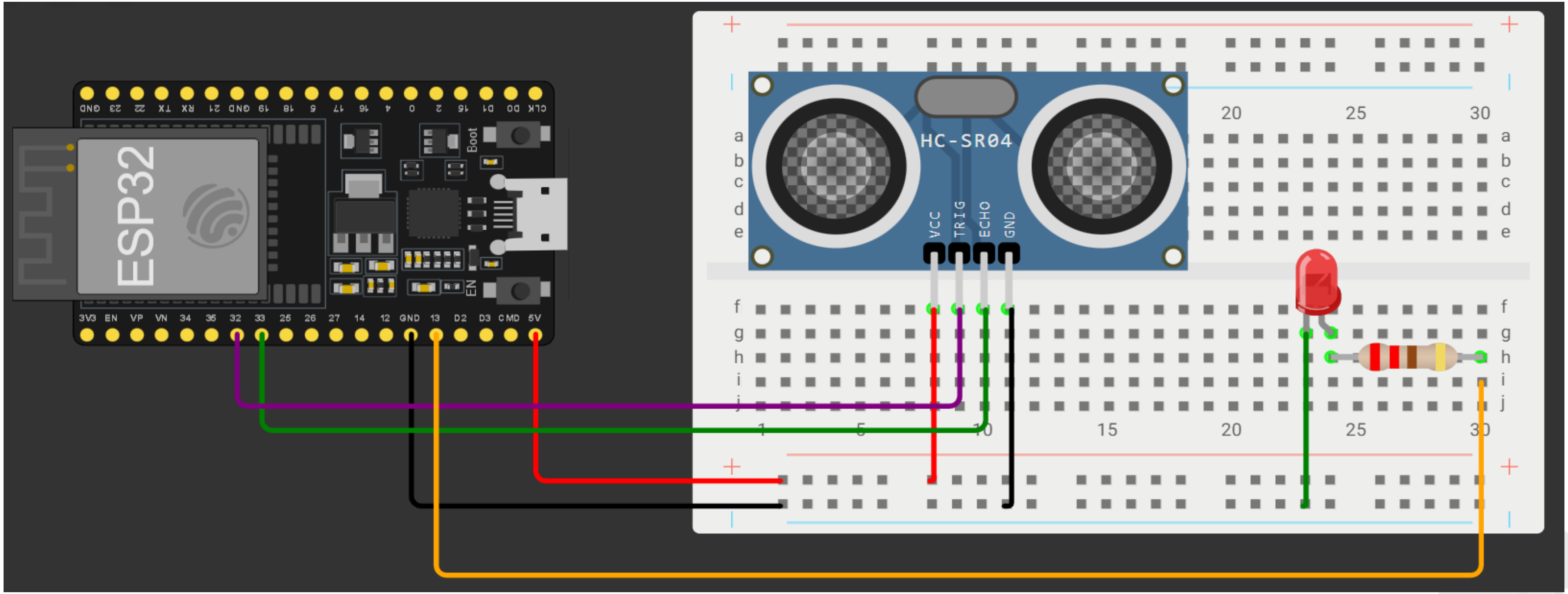
Ultrasonic Sensor with Led

Step 3: Connecting the Control Pins (TRIG and ECHO)



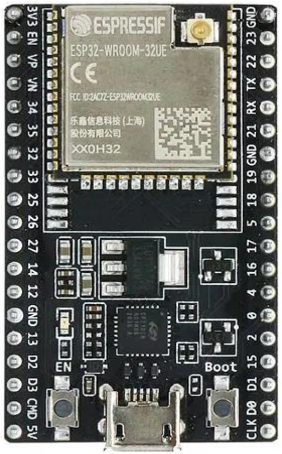
Ultrasonic Sensor with Led

Step 4: Connecting a Led



Firebase ESP32 Library

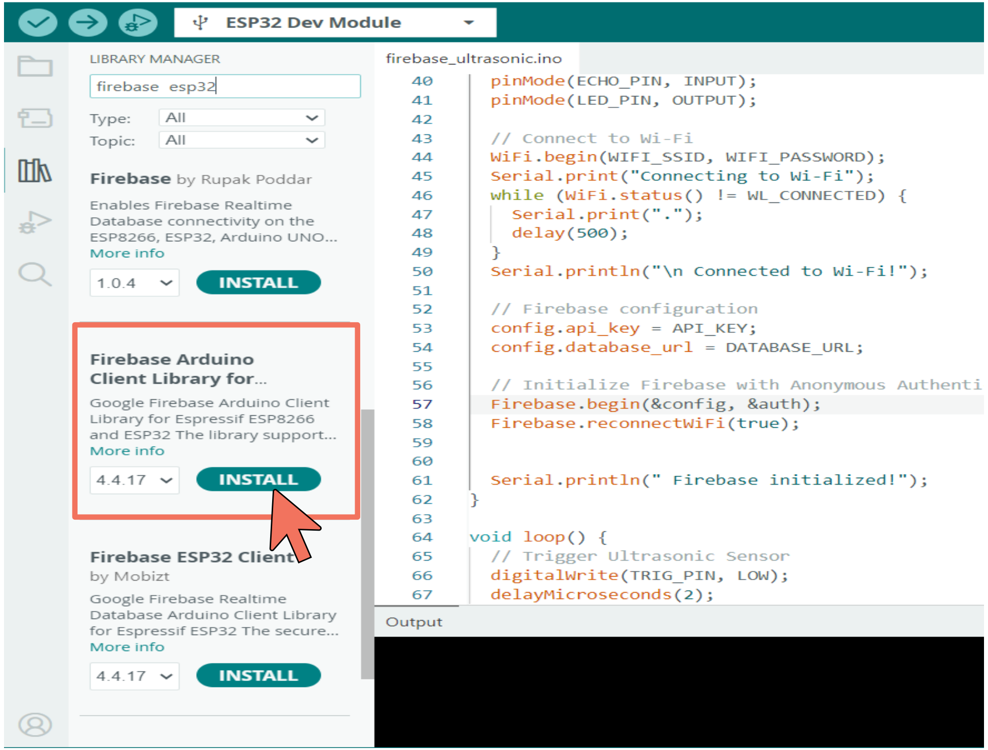
- The Firebase ESP32 library provides Firebase Realtime database and Firebase Cloud Messaging functions and supports ESP32.



Firebase

Firebase ESP32 Library

First, we need to install Firebase Library.



1. go to library manager from tools→ manage libraires and search for (firebase esp32)
2. Make sure to install the library shown in this picture.

base_ultrasonic | Arduino IDE 2.3.6

dit Sketch Tools Help

ESP32 Dev Module

firebase_ultrasonic.ino

```
#include <WiFi.h>                  // Library to connect ESP32 to Wi-Fi
#include <Firebase_ESP_Client.h>   // Library for Firebase functions
#include "addons/TokenHelper.h"    // Required helper for Firebase authentication
#include "addons/RTDBHelper.h"     // Required helper for Firebase Realtime Database

// Wi-Fi credentials
#define WIFI_SSID "rahmabadar"
#define WIFI_PASSWORD "rahma@011"

// Firebase credentials
#define API_KEY "AIzaSyALtMaCgDbtjnjoag8zCuQ7PzgEg5014SY"
#define DATABASE_URL "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/"
#define USER_EMAIL "rahma@gmail.com"
#define USER_PASSWORD "rahmaa123456"

// Pin Definitions
#define TRIG_PIN 32    // Trigger pin of ultrasonic sensor
#define ECHO_PIN 33    // Echo pin of ultrasonic sensor
#define LED_PIN 13     // Built-in LED pin

// Firebase Objects
FirebaseData fbdo;       // Object to handle Firebase data
FirebaseAuth auth;       // Object for Firebase authentication
FirebaseConfig config;   // Object for Firebase configuration

// Variables for distance measurement
long duration;
float distance;

```

firebase_ultrasonic | Arduino IDE 2.3.6

File Edit Sketch Tools Help

ESP32 Dev Module

firebase_ultrasonic.ino

```
void setup() {
  Serial.begin(115200); // Start serial communication for debugging

  // Initialize pins
  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);

  // Connect to Wi-Fi
  WiFi.begin(WIFI_SSID, WIFI_PASSWORD);
  Serial.print("Connecting to Wi-Fi");
  while (WiFi.status() != WL_CONNECTED) {
    Serial.print(".");
    delay(500);
  }
  Serial.println("\nConnected to Wi-Fi!");

  // Configure Firebase
  config.api_key = API_KEY;
  config.database_url = DATABASE_URL;

  auth.user.email = USER_EMAIL;
  auth.user.password = USER_PASSWORD;

  // Initialize Firebase connection
  Firebase.begin(&config, &auth);
  Firebase.reconnectWiFi(true); // Auto-reconnect Wi-Fi if disconnected

  Serial.println("Firebase initialized!");
}
```

firebase_ultrasonic | Arduino IDE 2.3.6

le Edit Sketch Tools Help

ESP32 Dev Module

firebase_ultrasonic.ino

```
void loop() {
  // Measure distance using ultrasonic sensor
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIG_PIN, LOW);

  duration = pulseIn(ECHO_PIN, HIGH);       // Measure the time for echo to return
  distance = duration * 0.034 / 2;          // Convert time to distance in cm

  Serial.print("Distance: ");
  Serial.println(distance);

  // Send distance to Firebase
  if (Firebase.RTDB.setFloat(&fbdo, "/distance", distance)) {
    Serial.println("Distance updated in Firebase");
  } else {
    Serial.print("Failed to send distance: ");
    Serial.println(fbdo.errorReason());
  }
  // Control LED based on distance
  if (distance < 10) {
    digitalWrite(LED_PIN, HIGH);              // Turn LED ON if object is closer than 10 cm
    Firebase.RTDB.setString(&fbdo, "/led", "ON");
  } else {
    digitalWrite(LED_PIN, LOW);               // Turn LED OFF otherwise
    Firebase.RTDB.setString(&fbdo, "/led", "OFF");
  }
  delay(2000); // Wait 2 seconds before next measurement
}
```

Ultrasonic Sensor with Led Data stored in firebase

firebase_ultrasonic.ino

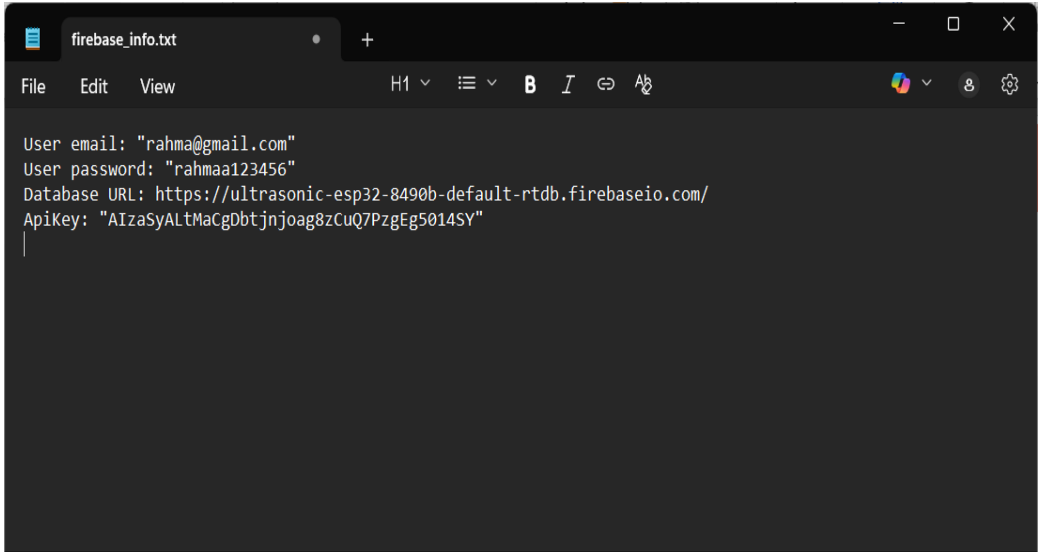
```
#include <WiFi.h>                 // Library to connect ESP32 to Wi-Fi
#include <Firebase_ESP_Client.h>  // Library for Firebase functions
#include "addons/TokenHelper.h"   // Required helper for Firebase authentication
#include "addons/RTDBHelper.h"    // Required helper for Firebase Realtime Database

// Wi-Fi credentials
#define WIFI_SSID "rahmabadar"
#define WIFI_PASSWORD "rahma@011"
```

Explanation:

- Here we enter the **Wi-Fi network name (SSID)**
- and **password** so that the **ESP32** can connect to the internet.
  This connection allows the board to send and receive data from **Firebase**

Ultrasonic Sensor with Led Data stored in firebase

```
// Firebase credentials
#define API_KEY "AIzaSyALtMaCgDbtjnjoag8zCuQ7PzgEg5014SY"
#define DATABASE_URL "https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/"
#define USER_EMAIL "rahma@gmail.com"
#define USER_PASSWORD "rahmaa123456"
```

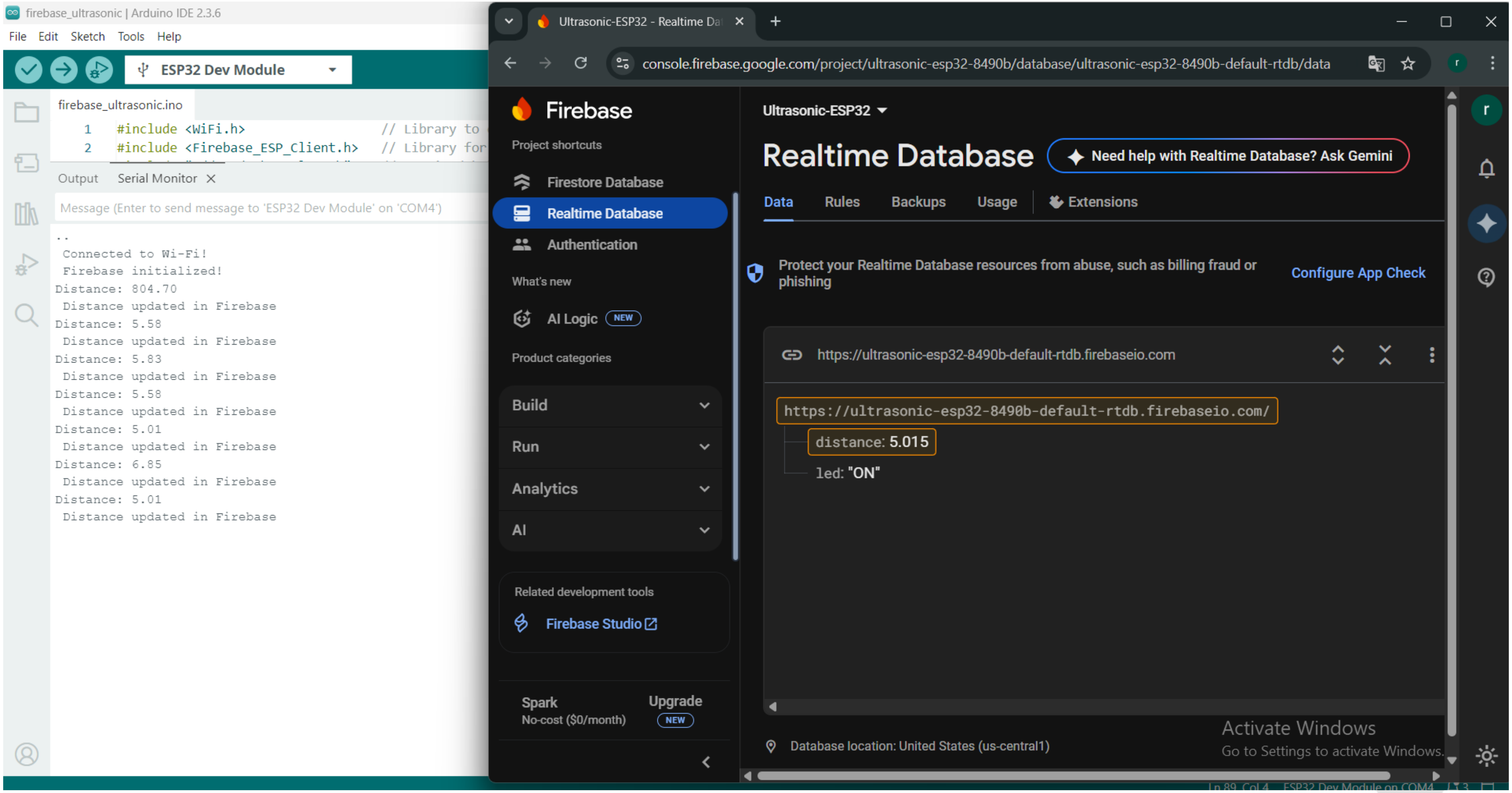


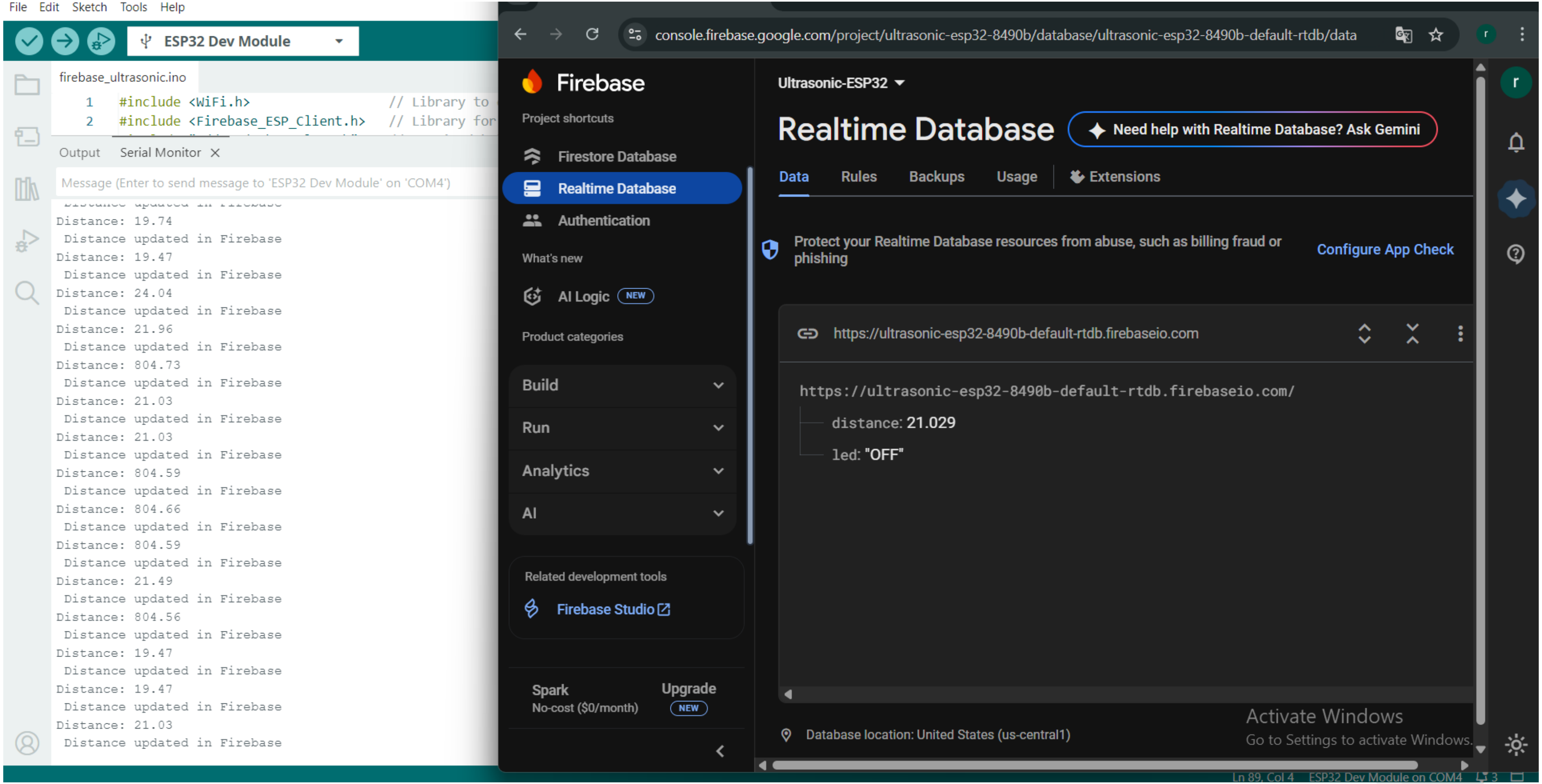
Explanation:

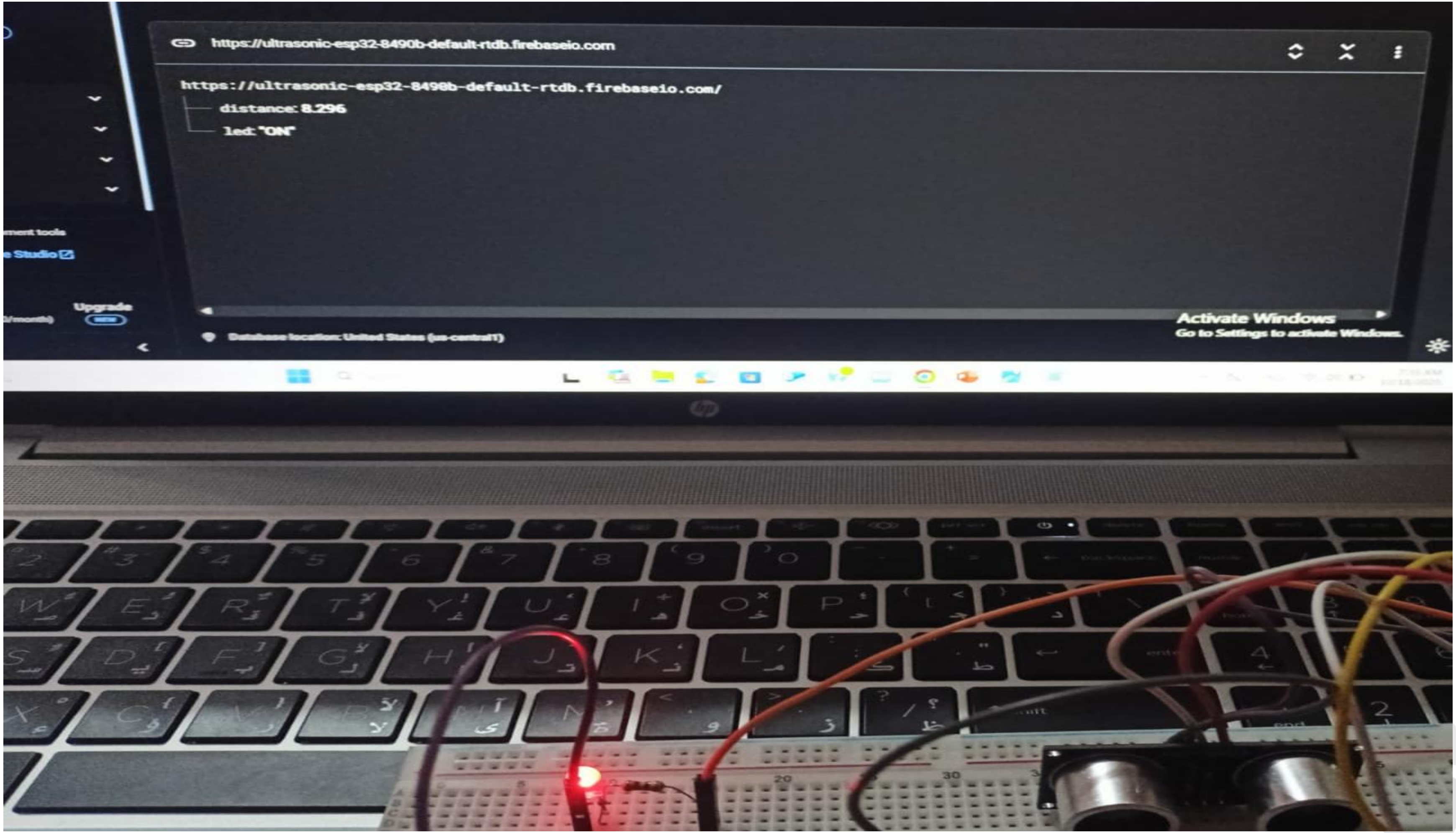
- Here is where we put the information we saved earlier in **Notepad** while setting up the project.

These include:

- **API_KEY** → the unique key for our Firebase project.
- **DATABASE_URL** → the address of our Realtime Database.
- **USER_EMAIL & USER_PASSWORD** → the login details we created for Firebase.


firebase_ultrasonic | Arduino IDE 2.3.6
File Edit Sketch Tools Help
ESP32 Dev Module
firebase_ultrasonic.ino
#include <WiFi.h> // Library to
#include <Firebase_ESP_Client.h> // Library for
Output Serial Monitor
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
..
Connected to Wi-Fi!
Firebase initialized!
Distance: 804.70
Distance updated in Firebase
Distance: 5.58
Distance updated in Firebase
Distance: 5.83
Distance updated in Firebase
Distance: 5.58
Distance updated in Firebase
Distance: 5.01
Distance updated in Firebase
Distance: 6.85
Distance updated in Firebase
Distance: 5.01
Distance updated in Firebase
Ultrasonic-ESP32 - Realtime Da
console.firebase.google.com/project/ultrasonic-esp32-8490b/database/ultrasonic-esp32-8490b-default-rtdb/data
Firebase
Project shortcuts
Firestore Database
Realtime Database
Authentication
What's new
AI Logic NEW
Product categories
Build
Run
Analytics
AI
Related development tools
Firebase Studio
Spark
No-cost ($0/month)
Upgrade
NEW
Ultrasonic-ESP32
Realtime Database
Need help with Realtime Database? Ask Gemini
Data Rules Backups Usage Extensions
Protect your Realtime Database resources from abuse, such as billing fraud or phishing
Configure App Check
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/
distance: 5.015
led: "ON"
Database location: United States (us-central1)
Activate Windows
Go to Settings to activate Windows.


File Edit Sketch Tools Help
ESP32 Dev Module
firebase_ultrasonic.ino
#include <WiFi.h> // Library to
#include <Firebase_ESP_Client.h> // Library for
Output
Serial Monitor
Message (Enter to send message to 'ESP32 Dev Module' on 'COM4')
Distance: 19.74
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 24.04
Distance updated in Firebase
Distance: 21.96
Distance updated in Firebase
Distance: 804.73
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase
Distance: 804.59
Distance updated in Firebase
Distance: 804.66
Distance updated in Firebase
Distance: 804.59
Distance updated in Firebase
Distance: 21.49
Distance updated in Firebase
Distance: 804.56
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 19.47
Distance updated in Firebase
Distance: 21.03
Distance updated in Firebase
console.firebase.google.com/project/ultrasonic-esp32-8490b/database/ultrasonic-esp32-8490b-default-rtdb/data
Firebase
Project shortcuts
Firestore Database
Realtime Database
Authentication
What's new
AI Logic NEW
Product categories
Build
Run
Analytics
AI
Related development tools
Firebase Studio
Spark
No-cost ($0/month)
Upgrade NEW
Ultrasonic-ESP32
Realtime Database
Need help with Realtime Database? Ask Gemini
Data
Rules
Backups
Usage
Extensions
Protect your Realtime Database resources from abuse, such as billing fraud or phishing
Configure App Check
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/
distance: 21.029
led: "OFF"
Database location: United States (us-central1)
Activate Windows
Go to Settings to activate Windows.
Ln 89, Col 4 ESP32 Dev Module on COM4


https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com
https://ultrasonic-esp32-8490b-default-rtdb.firebaseio.com/
distance: 8.296
led: "ON"
Database location: United States (us-central1)
Activate Windows
Go to Settings to activate Windows.

Thank you